

Foundational Mathematics for Water Engineering
MATH 128

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Welcome

Mathematics is a practical language for describing, measuring, and improving water systems. In Water Engineering Technology, that language appears in calculations involving flow, storage, treatment, pumping, chemical dosing, site geometry, and the interpretation of operational data. *Foundational Mathematics for Water Engineering* develops the mathematical habits needed to work carefully in those settings: stating assumptions, tracking units, representing relationships clearly, checking whether an answer is reasonable, and explaining what a result means.

This book is organized in six chapters. It begins with algebraic foundations and then develops functions, polynomial and rational models, exponential and logarithmic models, trigonometry, and descriptive statistics. Each chapter opens with learning outcomes, then introduces key terms in definition boxes immediately before they are needed. Concepts are developed through explanations, worked examples, applied problems, and practice. Chapter reviews include exercises with immediate feedback and open-ended questions for discussion.

The central perspective is that a calculation is not complete when a number appears on a calculator. A useful mathematical result includes a quantity, appropriate units, a sensible magnitude, and an interpretation in the setting that produced it. The examples use realistic but fictional water-engineering situations unless a cited source is necessary.

Learning goals

By the end of the book, readers should be able to:

- represent and solve quantitative water-engineering problems with appropriate algebra, functions, trigonometry, and descriptive statistics;
- interpret graphs, tables, formulas, and calculations in context;
- communicate assumptions, units, and conclusions clearly; and
- check the reasonableness of a calculation before using it to support a decision.

Software

Optional demonstrations use Desmos for graphing and Microsoft Excel for tables, calculations, and charts. Both are tools for checking and communicating mathematics, not substitutes for understanding the underlying method.

Chapter 1

The Basics

Water engineering calculations depend on reliable algebra. Flow, storage, concentration, pressure, and treatment problems all require careful work with numbers, expressions, equations, and inequalities. Following the sequence of the MATH 128 Fall 2026 notes, this chapter begins with real numbers and arithmetic, then develops sets, exponents, roots, polynomials, equations, lines, rational expressions, and inequalities. Units and physical restrictions remain part of every calculation. A solution is complete only when its algebra is correct, its units are appropriate, and its meaning is reasonable in context.

Learning outcomes. By the end of this chapter, you should be able to:

- apply real-number, fraction, exponent, and root properties;
- represent sets and ranges using inequalities and interval notation;
- simplify, expand, factor, and divide algebraic expressions;
- apply the Remainder, Factor, and Rational Zeros Theorems;
- solve and check linear, quadratic, radical, and formula-based equations;
- determine equations of lines and interpret slope; and
- solve linear, polynomial, rational, and absolute-value inequalities.

1.1 Real numbers and their properties

Measurements, counts, calculated values, and model outputs all belong to a common number system.

Real number. A number represented by a point on the real number line. The set of all real numbers is denoted by $\mathbb{R}$.

Rational number. A number that can be written as a/b , where a and b are integers and $b \neq 0$.

The numbers used in measurement and calculation are real numbers, denoted by $\mathbb{R}$. They include natural numbers, whole numbers, integers, rational numbers, and irrational numbers. A rational number can be written as a/b , where a and b are integers and $b \neq 0$. Irrational numbers, including $\sqrt{2}$ and π , cannot be written in that form.

For real numbers a , b , and c , addition and multiplication are commutative and associative. The distributive property is

$$a(b + c) = ab + ac.$$

The commutative property changes order, while the associative property changes grouping. Neither property permits arbitrary rearrangement of subtraction or division. For instance, $8 - 3$ is not equal to $3 - 8$. The properties of negatives follow from multiplication by -1 :

$$(-1)a = -a, \quad -(-a) = a, \quad (-a)(-b) = ab,$$

$$-(a + b) = -a - b, \quad -(a - b) = -a + b.$$

These identities are especially important when a negative sign appears outside a long formula. A reliable practice is to regard the sign as a factor of -1 and distribute it explicitly.

1.1.1 Translating signs carefully

The negative sign can indicate a negative number, subtraction, or multiplication by -1 . These meanings are related but should not be blurred. In -3^2 , the exponent applies before the negative sign, so the value is $-(3^2) = -9$. In $(-3)^2$, the parentheses make -3 the base, so the value is 9 . Parentheses are therefore essential when a negative value is substituted into a power.

Engineering sign conventions should be stated before calculation. A negative elevation can mean below a chosen datum. A negative change in tank depth can mean that depth decreased. Neither represents a negative physical length. The sign communicates direction relative to a reference.

Worked Example: Evaluating a negative squared correction

A calibration calculation contains -0.4^2 . Because the exponent is evaluated first,

$$-0.4^2 = -(0.4^2) = -0.16.$$

If the intended base were -0.4 , the expression would need to be written $(-0.4)^2 = 0.16$.

Try It. Evaluate -1.5^2 and $(-1.5)^2$.

Check Your Work

$-1.5^2 = -2.25$, while $(-1.5)^2 = 2.25$.

Worked Example: Applying the distributive property

Simplify $3(8 - 2y)$:

$$3(8 - 2y) = 24 - 6y.$$

Every term inside the parentheses is multiplied by 3 .

Try It. Simplify $-4(2 - 3x)$.

Check Your Work

$-8 + 12x$.

Worked Example: Classifying a measured value

A recorded turbidity reading of 2.75 can be written as $275/100 = 11/4$. It is therefore rational and real. It is not an integer.

Try It. Classify -6 , $\sqrt{7}$, and $4/9$ using the smallest applicable number set.

Check Your Work

-6 is an integer, $\sqrt{7}$ is irrational, and $4/9$ is rational.

Worked Example: Interpreting a signed water-level change

A reservoir level changes by -0.18 m in the morning and by $+0.07$ m in the afternoon. The net change is

$$-0.18 + 0.07 = -0.11 \text{ m.}$$

The negative result means that the final level is 0.11 m below the initial level.

Try It. A wet well level changes by -0.24 m and then by $+0.31$ m. Find and interpret the net change.

Check Your Work

$-0.24 + 0.31 = 0.07$ m, so the final level is 0.07 m above the initial level.

Applied Problem: Tracking a reservoir level. A reservoir begins the day at 6.42 m relative to its operating datum. Withdrawal lowers the level by 0.38 m, inflow raises it by 0.21 m, and evaporation lowers it by another 0.03 m. Determine the final level and the net change.

Check Your Work

The final level is $6.42 - 0.38 + 0.21 - 0.03 = 6.22$ m. The net change is $6.22 - 6.42 = -0.20$ m, so the reservoir finishes 0.20 m below its starting level.

1.1.2 Practice Problems

1. Simplify $5(3 - 2x)$.

Check Your Work

$$15 - 10x.$$

2. Name the smallest applicable set for 0.

Check Your Work

The whole numbers.

3. Is 0.125 rational? Explain.

Check Your Work

Yes. $0.125 = 1/8$.

4. Simplify $-(7 - y)$.

Check Your Work

$$-7 + y.$$

5. A measured flow is 12.4 L/s. Identify the smallest familiar number set containing 12.4.

Check Your Work

It is rational because $12.4 = 62/5$.

6. A tank level falls 0.35 m and then rises 0.12 m. Find the net change.

Check Your Work

$-0.35 + 0.12 = -0.23$ m. The level falls by 0.23 m overall.

7. A pressure correction is written $-(1.8 - 0.6)$. Simplify it.

Check Your Work

$$-(1.8 - 0.6) = -1.2.$$

8. A flow measurement is $\sqrt{50}$ L/s. Is the exact value rational or irrational?

Check Your Work

$\sqrt{50} = 5\sqrt{2}$, which is irrational.

1.2 BEDMAS and fractions

Reliable calculations require an agreed order for interpreting operations and a consistent way to work with parts of a whole.

Fraction. A quotient a/b in which a is the numerator, b is the denominator, and $b \neq 0$.

BEDMAS records the order of operations: brackets, exponents, division and multiplication from left to right, then addition and subtraction from left to right. Fractions follow the same rules. To multiply fractions, multiply their numerators and denominators. To divide, multiply by the reciprocal. To add or subtract, first use a common denominator.

For nonzero denominators,

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \quad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c},$$

and

$$\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}.$$

When denominators differ, use the least common denominator rather than adding denominators. A fraction bar also acts as a grouping symbol. Everything in its numerator is evaluated as one group and everything in its denominator as another. In an engineering calculation, keep several digits during intermediate work and round only the final result to a precision justified by the data.

1.2.1 Complex numerical fractions

A complex fraction has one or more fractions in its numerator or denominator. Simplify the numerator and denominator separately, or multiply both by the least common denominator of all smaller fractions. For example,

$$\frac{\frac{1}{6} + \frac{4}{9}}{\frac{2}{4} - \frac{2}{10}} = \frac{\frac{11}{18}}{\frac{3}{10}} = \frac{11}{18} \cdot \frac{10}{3} = \frac{55}{27}.$$

The equality at every stage makes the order visible. Entering an entire complex fraction into a calculator without grouping parentheses can produce a different expression.

Worked Example: Comparing two fractional operating periods

A pump operates for $3/8$ of an hour at one setting and $5/12$ of an hour at another. The total operating time is

$$\frac{3}{8} + \frac{5}{12} = \frac{9}{24} + \frac{10}{24} = \frac{19}{24} \text{ h.}$$

Try It. A backwash cycle uses $2/15$ h for draining and $1/10$ h for rinsing. Find the combined time.

Check Your Work

$$2/15 + 1/10 = 4/30 + 2/30 = 6/30 \text{ h, or 12 min.}$$

Worked Example: Combining fractions of capacity

If two operating periods process $1/6$ and $4/9$ of a tank capacity, then

$$\frac{1}{6} + \frac{4}{9} = \frac{3}{18} + \frac{8}{18} = \frac{11}{18}.$$

Try It. Combine $1/4$ and $2/3$ of a tank capacity.

Check Your Work

$$1/4 + 2/3 = 3/12 + 8/12 = 11/12.$$

1.2.2 Units, conversion factors, and precision

Units may be treated as algebraic factors. A conversion factor is a ratio equal to 1, such as $1000 \text{ L}/1 \text{ m}^3$. Arrange it so the unwanted unit cancels. For example,

$$2.4 \text{ m}^3 \left(\frac{1000 \text{ L}}{1 \text{ m}^3} \right) = 2400 \text{ L}.$$

This cancellation shows why multiplying by 1000 is appropriate. If the units do not cancel as intended, the conversion factor is upside down or the relationship is incomplete. A rate such as litres per minute is a fraction, so multiplying by minutes produces litres.

Exact numbers, such as 1000 L in 1 m^3 , do not limit precision. Measured values do. Retain guard digits during a calculation and round the final result once. A reported answer should not imply more precision than the measurements support. Estimation provides a quick reasonableness check: a basin roughly 20 m by 8 m by 3 m should have a volume near 480 m^3 , not 48 m^3 or 4800 m^3 .

Worked Example: Calculating a rectangular basin volume

A clearwell is 18.0 m long, 7.5 m wide, and filled to a depth of 3.2 m . Using $V = LWD$,

$$V = (18.0)(7.5)(3.2) = 432 \text{ m}^3.$$

The numerical result is 432, and its unit is cubic metres because three lengths were multiplied.

Try It. Find the volume of a basin measuring 12 m by 6.5 m by 2.0 m .

Check Your Work

$$V = (12)(6.5)(2.0) = 156 \text{ m}^3.$$

Worked Example: Converting a daily flow to litres

A small facility treats 0.85 m^3 of water per day. Since $1 \text{ m}^3 = 1000 \text{ L}$,

$$0.85 \text{ m}^3 \left(\frac{1000 \text{ L}}{1 \text{ m}^3} \right) = 850 \text{ L}.$$

Try It. Convert 3.6 m^3 to litres.

Check Your Work

$$3.6(1000) = 3600 \text{ L}.$$

Applied Problem: Estimating stored water. A rectangular equalization basin is 14.0 m long and 6.0 m wide. The water depth is $2\frac{1}{4}$ m. Find the stored volume in cubic metres and litres.

Check Your Work

Convert $2\frac{1}{4}$ to 2.25. Then $V = (14.0)(6.0)(2.25) = 189 \text{ m}^3$. Multiplying by 1000 gives 189,000 L.

1.2.3 Practice Problems

- Evaluate $18 - 3(4 + 1)$.
Check Your Work
 $18 - 15 = 3$.
- Evaluate $6 + 2^3(5 - 2)$.
Check Your Work
 $6 + 8(3) = 30$.
- Calculate $3/8 + 5/12$.
Check Your Work
 $9/24 + 10/24 = 19/24$.
- Calculate $(7/10) \div (14/15)$.
Check Your Work
 $(7/10)(15/14) = 3/4$.
- A tank is 9.0 m by 4.0 m by 2.5 m. Find its volume.
Check Your Work
 $V = (9.0)(4.0)(2.5) = 90 \text{ m}^3$.
- A filter is online for $5/12$ of a day and in standby for $1/8$ of a day. What fraction of the day do these periods occupy?
Check Your Work
 $5/12 + 1/8 = 10/24 + 3/24 = 13/24$ of a day.
- Convert 4.75 m^3 of water to litres.
Check Your Work
 $4.75(1000) = 4750 \text{ L}$.
- A chemical feed pump delivers 18 L/min for 35 min. Find the delivered volume.
Check Your Work
 $(18 \text{ L/min})(35 \text{ min}) = 630 \text{ L}$.

1.3 Sets, inequalities, and interval notation

Operating conditions are often described as collections of permitted values rather than as a single measurement.

Set. A collection of distinct objects called elements.

Interval. A continuous set of real numbers between specified endpoints.

A set is written with curly braces, as in $A = \{2, 4, 6\}$. The intersection $A \cap B$ contains elements common to both sets. The union $A \cup B$ contains elements in either set or both.

Set-builder notation describes sets of real numbers, such as

$$\{x \in \mathbb{R} \mid -3 \leq x < -1\}.$$

The same set is $[-3, -1)$ in interval notation. A square bracket includes an endpoint. A parenthesis excludes

it. Infinity always receives a parenthesis. Brackets are used here because they communicate endpoint inclusion, not because the endpoints are numbers.

The order symbols have geometric meaning. If a lies to the left of b on the number line, then $a < b$ and $b > a$. The symbols $\leq$ and $\geq$ include equality. Set-builder notation states both the membership rule and, when needed, the number system. Thus $\{x \in \mathbb{R} \mid x \geq 5\}$ and $[5, \infty)$ describe the same real-number set.

Intersections often model simultaneous requirements. If a pump must satisfy both an equipment limit and a process limit, its permitted settings lie in the intersection of the two ranges. Unions model alternatives, where satisfying either range is sufficient. Sketching both sets on one number line is an effective way to verify an interval calculation.

Worked Example: Combining inspection sets

If $A = \{1, 3, 5, 7\}$ and $B = \{3, 4, 5, 6\}$, then $A \cap B = \{3, 5\}$ and $A \cup B = \{1, 3, 4, 5, 6, 7\}$.

Try It. Find the intersection of $C = \{2, 4, 6\}$ and $D = \{1, 2, 3, 4\}$.

Check Your Work

$C \cap D = \{2, 4\}$.

Worked Example: Intersecting two operating limits

A pump is rated for $20 \leq q \leq 42$ L/s, while a process requires $28 \leq q \leq 50$ L/s. Both conditions hold on their intersection:

$$[20, 42] \cap [28, 50] = [28, 42].$$

Try It. Find the intersection of $[15, 35]$ and $[24, 40]$.

Check Your Work

$[24, 35]$.

1.3.1 Converting among representations

To move from an inequality to a graph, first locate each endpoint, decide whether it is included, and shade the permitted side or region. Use a solid point for an included endpoint and an open point for an excluded endpoint. Interval notation records the same distinction with a square bracket or parenthesis.

For example, $x \geq -3$ begins at -3 , includes that endpoint, and extends without bound to the right. Its interval is $[-3, \infty)$. The infinity symbol receives a parenthesis because infinity is not a real endpoint. For a bounded condition such as $-4 < x \leq 2.5$, the interval is $(-4, 2.5]$.

Worked Example: Recording a flow range

A filter may operate from 18 L/s through 26 L/s, including both limits. If q is flow rate, then

$$18 \leq q \leq 26.$$

The corresponding interval is $[18, 26]$.

Try It. Write the range from 12 L/s, excluded, through 20 L/s, included, as an inequality and interval.

Check Your Work

$12 < q \leq 20$ and $(12, 20]$.

Worked Example: Converting a chlorine-residual specification

A specification requires a residual greater than 0.20 mg/L but no greater than 0.80 mg/L. If C is the residual,

$$0.20 < C \leq 0.80,$$

which is $(0.20, 0.80]$ in interval notation.

Try It. Express $6.5 \leq p < 8.5$ in interval notation.

Check Your Work

$[6.5, 8.5)$.

Applied Problem: Selecting an allowable filter loading rate. A filter is mechanically rated from 4 through 12 m/h. Water-quality conditions require a rate greater than 6 m/h but less than 10 m/h. Write each range in interval notation and determine the allowable intersection.

Check Your Work

The equipment range is $[4, 12]$ and the process range is $(6, 10)$. Their intersection is $(6, 10)$ m/h.

1.3.2 Practice Problems

- Write $x \geq -3$ in interval notation.
Check Your Work
 $[-3, \infty)$.
- Write $(-4, 7)$ as an inequality.
Check Your Work
 $-4 < x < 7$.
- Find $\{1, 2, 5\} \cap \{2, 3, 5\}$.
Check Your Work
 $\{2, 5\}$.
- Find $\{1, 2\} \cup \{2, 4\}$.
Check Your Work
 $\{1, 2, 4\}$.
- A pressure must be below 550 kPa. Express the condition for p .
Check Your Work
 $p < 550$, or $(-\infty, 550)$ algebraically.
- A tank level must satisfy $2.0 \leq h \leq 4.5$ m. Write the interval.
Check Your Work
 $[2.0, 4.5]$ m.
- A pump can provide $[18, 44]$ L/s, but a process requires $[25, 38]$ L/s. Find the common operating range.
Check Your Work
 $[18, 44] \cap [25, 38] = [25, 38]$ L/s.
- A turbidity alarm is triggered below 0.1 NTU or above 1.0 NTU. Express the alarm set as a union of intervals.
Check Your Work
 $(-\infty, 0.1) \cup (1.0, \infty)$.

1.4 Absolute value and distance

Signed measurements describe direction relative to a reference, while many engineering questions require only the magnitude of a difference.

Absolute value. The distance of a real number from zero on the number line.

Absolute value measures distance from zero, so $|a| \geq 0$. The distance between a and b is

$$d(a, b) = |b - a|.$$

More formally,

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

Other useful properties are $|ab| = |a||b|$, $|a/b| = |a|/|b|$ for $b \neq 0$, and the triangle inequality $|a + b| \leq |a| + |b|$. Absolute value reports magnitude without direction. A signed deviation can show whether a reading is above or below target, while its absolute value shows how far it is from target.

Absolute value is not an instruction to remove a negative sign mechanically. Evaluate the expression inside the bars first, then take its distance from zero. Thus $|3 - 8| = |-5| = 5$. The order inside the bars matters, but the distance between two values does not: $|b - a| = |a - b|$.

For elevations -2.4 m and 5.1 m, the distance is

$$|5.1 - (-2.4)| = 7.5 \text{ m.}$$

Worked Example: Distance between survey readings

Two readings relative to a benchmark are -2.4 m and 5.1 m. Their distance is

$$d = |5.1 - (-2.4)| = 7.5 \text{ m.}$$

Distance is positive even though one position is below the benchmark.

Try It. Find the distance between -3.2 m and 1.9 m.

Check Your Work

$$|1.9 - (-3.2)| = 5.1 \text{ m.}$$

Worked Example: Interpreting a deviation

A sensor reads 0.8 units below its reference, so its signed deviation is -0.8 . The magnitude of the deviation is $|-0.8| = 0.8$ units.

Try It. A reading differs from its target by -1.25 mg/L. Find the magnitude of the difference.

Check Your Work

$$|-1.25| = 1.25 \text{ mg/L.}$$

Worked Example: Measuring departure from a pH target

A process target is pH 7.20 and the measured value is 6.85 . The signed deviation is $6.85 - 7.20 = -0.35$, while the magnitude of the deviation is

$$|6.85 - 7.20| = 0.35.$$

Try It. Find the absolute deviation of a pH reading of 7.62 from a target of 7.25.

Check Your Work

$$|7.62 - 7.25| = 0.37.$$

Worked Example: Comparing two pipe elevations

The invert elevations of two pipe ends are 412.84 m and 410.19 m. Their vertical separation is

$$|410.19 - 412.84| = 2.65 \text{ m.}$$

Try It. Find the separation between elevations 506.30 m and 503.75 m.

Check Your Work

$$|503.75 - 506.30| = 2.55 \text{ m.}$$

Applied Problem: Checking a chemical-feed tolerance. A controller is set to deliver 24.0 L/h. During four inspections, the measured rates are 23.4, 24.3, 25.1, and 23.8 L/h. Calculate the absolute deviation from the set point for each reading and identify the largest deviation.

Check Your Work

The absolute deviations are 0.6, 0.3, 1.1, and 0.2 L/h. The largest deviation is 1.1 L/h, produced by the 25.1 L/h reading.

1.4.1 Practice Problems

- Evaluate $|-12|$.
Check Your Work
12.
- Evaluate $|4 - 11|$.
Check Your Work
7.
- Find the distance between -6 and 9 .
Check Your Work
15.
- Solve $|x| = 5$.
Check Your Work
 $x = -5$ or $x = 5$.
- Two elevations are 102.4 m and 98.7 m. Find their separation.
Check Your Work
3.7 m.
- A chlorine residual is 0.18 mg/L below a 0.75 mg/L target. Find the magnitude of the deviation.
Check Your Work
0.18 mg/L.
- Find the vertical separation between pipe elevations 284.65 m and 281.90 m.
Check Your Work
 $|281.90 - 284.65| = 2.75 \text{ m.}$
- A flow reading is 42.6 L/s and its reference value is 45.0 L/s. Find the signed deviation and absolute deviation.
Check Your Work
The signed deviation is $42.6 - 45.0 = -2.4 \text{ L/s}$, and the absolute deviation is 2.4 L/s.

1.5 Exponents, roots, and rational exponents

Repeated multiplication and its inverse operations provide compact ways to express area, volume, scaling, and scientific notation.

Exponent. A number that indicates the power to which a base is raised.

Root. A value that produces a given number when raised to a specified power.

The exponent laws include

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}.$$

Products and quotients raised to a power satisfy

$$(ab)^n = a^n b^n, \quad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

These rules apply to multiplication and division, not addition. In general, $(a + b)^n \neq a^n + b^n$. Negative exponents do not make a value negative. They indicate reciprocals.

For $a \neq 0$, $a^0 = 1$ and $a^{-n} = 1/a^n$. Rational exponents connect powers and roots:

$$a^{m/n} = \sqrt[n]{a^m}.$$

When n is even, a real principal root requires a nonnegative radicand, and $\sqrt{x^2} = |x|$. To rationalize $5/\sqrt{3}$, multiply by $\sqrt{3}/\sqrt{3}$ to obtain $5\sqrt{3}/3$.

Root properties parallel exponent properties when all expressions are defined:

$$\sqrt[n]{ab} = \sqrt[n]{a} \sqrt[n]{b}, \quad \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}.$$

For odd n , $\sqrt[n]{a^n} = a$. For even n , $\sqrt[n]{a^n} = |a|$. When a denominator has the form $A + B\sqrt{C}$, multiply by its conjugate $A - B\sqrt{C}$. Their product is $A^2 - B^2C$, which contains no square root.

Worked Example: Simplifying exponent notation

$$\frac{x^7 x^{-2}}{x^3} = x^{7-2-3} = x^2, \quad x \neq 0.$$

Try It. Simplify $(a^3)^2/a^4$.

Check Your Work
 a^2 , with $a \neq 0$.

Worked Example: Writing a small concentration in scientific notation

A concentration of 0.000042 g/L can be written

$$0.000042 \text{ g/L} = 4.2 \times 10^{-5} \text{ g/L}.$$

Try It. Write 0.0000068 g/L in scientific notation.

Check Your Work

$$6.8 \times 10^{-6} \text{ g/L}.$$

1.5.1 Simplifying roots

Look for perfect-power factors before approximating a root. For example,

$$\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}.$$

An exact radical form retains full precision and often simplifies later algebra. A decimal approximation may be added when the application requires a numerical measurement. When variables appear under an even root, absolute value may be required. For instance, $\sqrt{49x^2} = 7|x|$ for real x .

The conjugate method follows the difference-of-squares identity. Multiplying $2/(7+\sqrt{3})$ by $(7-\sqrt{3})/(7-\sqrt{3})$ produces denominator $49-3=46$ and numerator $14-2\sqrt{3}$. The simplified result is $(7-\sqrt{3})/23$.

Worked Example: Rationalizing a denominator

$$\frac{5}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{5\sqrt{3}}{3}.$$

Multiplying by $\sqrt{3}/\sqrt{3}$ changes the form but not the value.

Try It. Rationalize $4/\sqrt{5}$.

Check Your Work

$$4\sqrt{5}/5.$$

Worked Example: Finding the side of a square settling basin

A square basin has surface area 180 m^2 . If its side length is s , then $s^2 = 180$, so

$$s = \sqrt{180} = 6\sqrt{5} \approx 13.4 \text{ m}.$$

Only the positive root is meaningful as a length.

Try It. Find the positive side length of a square basin with area 98 m^2 .

Check Your Work

$$s = \sqrt{98} = 7\sqrt{2} \approx 9.90 \text{ m}.$$

Applied Problem: Scaling storage capacity. Two geometrically similar tanks have linear scale factor 1.5. If the smaller tank holds 32 m^3 , use the cubic scale factor to estimate the larger tank's capacity.

Check Your Work

Volume scales with the cube of the linear factor. The capacity is $32(1.5)^3 = 32(3.375) = 108 \text{ m}^3$.

1.5.2 Practice Problems

1. Simplify y^4y^3 .
Check Your Work
 y^7 .
2. Rewrite z^{-3} without a negative exponent.
Check Your Work
 $1/z^3, z \neq 0$.
3. Write $\sqrt[3]{x^2}$ with a rational exponent.
Check Your Work
 $x^{2/3}$.
4. Simplify $\sqrt{49a^2}$.
Check Your Work
 $7|a|$.
5. Rationalize $3/\sqrt{2}$.
Check Your Work
 $3\sqrt{2}/2$.
6. A bacterial count is 3.4×10^5 cells/mL. Write the count in ordinary notation.
Check Your Work
340,000 cells/mL.
7. A square tank cover has area 72 m^2 . Give its exact side length in simplified radical form.
Check Your Work
 $\sqrt{72} = 6\sqrt{2} \text{ m}$.
8. A cylindrical tank's dimensions are doubled. By what factor does its volume change?
Check Your Work
Volume changes by $2^3 = 8$, so it is eight times as large.

1.6 Algebraic expressions and polynomials

Engineering formulas frequently combine several variable terms into a single expression that can be evaluated or simplified.

Polynomial. A finite sum of terms whose variable exponents are nonnegative integers.

A polynomial in x has the form

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where its exponents are nonnegative integers. Polynomials are defined for every real input. Like terms may be combined, and products are expanded using the distributive property. For a channel with width $x + 3$ metres and depth $x - 1$ metres,

$$A = (x + 3)(x - 1) = x^2 + 2x - 3,$$

with the physical restriction $x > 1$.

The coefficient of the highest-degree term is the leading coefficient. A constant nonzero polynomial has degree zero. In two variables, the degree of a term is the sum of its exponents, so $4x^2y^3$ has degree 5. Expressions containing variables in denominators, negative variable exponents, or variable square roots are algebraic expressions but are not polynomials.

Domain is determined before algebraic manipulation. Polynomials accept every real input. A denominator cannot be zero. An even root requires a nonnegative radicand, and an even root in a denominator requires

a strictly positive radicand. These rules will recur when functions are introduced.

The special-product identities provide efficient expansion:

$$(A + B)^2 = A^2 + 2AB + B^2,$$

$$(A - B)^2 = A^2 - 2AB + B^2,$$

$$(A + B)(A - B) = A^2 - B^2.$$

Worked Example: Expanding a channel-area expression

For width $x + 3$ metres and depth $x - 1$ metres,

$$A = (x + 3)(x - 1) = x^2 + 2x - 3.$$

The physical setting also requires $x > 1$.

Try It. Expand the area of a channel with width $x + 4$ and depth $x - 2$.

Check Your Work

$x^2 + 2x - 8$, with $x > 2$ physically.

Worked Example: Combining polynomial terms

$$3x^3 - 9x - 2x^3 + 4x + 1 = x^3 - 5x + 1.$$

Try It. Simplify $7t^2 - 3t + 2t^2 + 5t$.

Check Your Work

$9t^2 + 2t$.

1.6.1 Polynomials in two variables

The expression $5x^2y - 3xy^3 + 7$ is a polynomial in x and y . The term $5x^2y$ has degree 3, the term $-3xy^3$ has degree 4, and the constant has degree 0. The polynomial therefore has degree 4. Like terms must have the same variables raised to the same powers. Thus $3x^2y$ and $-8x^2y$ combine, but x^2y and xy^2 do not.

When multiplying polynomials, every term in one factor multiplies every term in the other. FOIL is a memory aid for two binomials, but the distributive property is the general rule. Organizing products by descending degree helps reveal like terms and makes the result easier to check.

Worked Example: Combining a two-variable flow model

Suppose two contributions to a flow model are $4q^2t - 3qt^2$ and $-q^2t + 5qt^2$. Combining like terms gives

$$(4q^2t - q^2t) + (-3qt^2 + 5qt^2) = 3q^2t + 2qt^2.$$

Try It. Simplify $(6x^2y - 2xy^2) + (3x^2y + 5xy^2)$.

Check Your Work

$9x^2y + 3xy^2$.

1.6.2 Domains of algebraic expressions

An algebraic expression can contain roots, variable denominators, and noninteger powers. Its domain is the set of real inputs for which every operation is defined. For $\sqrt{5-x}/(x+9)$, the square root requires $x \leq 5$ and the denominator requires $x \neq -9$. Both conditions must hold, so the domain is $(-\infty, -9) \cup (-9, 5]$.

For $\sqrt{2-x}/\sqrt{x+3}$, the numerator requires $x \leq 2$. Because the square root is in the denominator, its radicand must be strictly positive, giving $x > -3$. The combined domain is $(-3, 2]$. Writing each restriction separately before taking their intersection makes complicated domains much easier to determine.

Expressions should normally be simplified only after the domain is known. A cancelled factor may remove the visible source of a restriction, but it cannot add a value that was undefined in the original expression.

Worked Example: Determining the domain of a depth formula

Consider $D(q) = \sqrt{40-q}/(q-8)$. The square root requires $q \leq 40$, and the denominator requires $q \neq 8$. Therefore,

$$\text{domain}(D) = (-\infty, 8) \cup (8, 40].$$

Try It. Find the domain of $\sqrt{25-x}/(x+5)$.

Check Your Work

$x \leq 25$ and $x \neq -5$, so the domain is $(-\infty, -5) \cup (-5, 25]$.

Applied Problem: Modelling a trapezoidal channel area. A channel has bottom width b , water depth d , and side slopes that add $d/2$ metres of width on each side. Write and expand a polynomial for the cross-sectional area using average width times depth. Then evaluate it at $b = 3.0$ m and $d = 1.6$ m.

Check Your Work

The top width is $b+d$, so the average width is $[b+(b+d)]/2 = b+d/2$. Thus $A = d(b+d/2) = bd + d^2/2$. At $b = 3.0$ and $d = 1.6$, $A = (3.0)(1.6) + (1.6)^2/2 = 6.08$ m².

1.6.3 Practice Problems

- State the degree of $4x^5 - 2x + 1$.
Check Your Work
5.
- Simplify $5x^2 + 3x - 2x^2 + x$.
Check Your Work
 $3x^2 + 4x$.
- Expand $(x+2)(x+5)$.
Check Your Work
 $x^2 + 7x + 10$.
- Expand $(2x-3)^2$.
Check Your Work
 $4x^2 - 12x + 9$.
- State the physical domain for area $A = x(x-1)$ when both dimensions are lengths.
Check Your Work
 $x > 1$.
- A rectangular channel has width $d+4$ and depth d . Expand its area formula.
Check Your Work
 $A = d(d+4) = d^2 + 4d$.
- Simplify the combined flow-model terms $7q^2t - 2qt^2 - 3q^2t + 6qt^2$.

Check Your Work

$$4q^2t + 4qt^2.$$

8. Find the physical domain of $C(v) = \sqrt{60 - v}/(v - 10)$ when v is nonnegative.

Check Your Work

$$0 \leq v \leq 60 \text{ with } v \neq 10, \text{ or } [0, 10) \cup (10, 60].$$

1.7 Expanding and factoring

The form of an expression affects which features are easy to see and which operations are easy to perform.

Factor. An expression multiplied by another expression to produce a given product.

Factoring reverses expansion. Begin with the greatest common factor, then look for patterns such as

$$A^2 - B^2 = (A + B)(A - B).$$

Always check a factorization by multiplying the factors. A complete factorization removes the greatest common factor first and continues until none of the remaining factors can be factored over the number system in use. The perfect-square patterns are

$$A^2 + 2AB + B^2 = (A + B)^2,$$

$$A^2 - 2AB + B^2 = (A - B)^2.$$

The sum and difference of cubes are

$$A^3 + B^3 = (A + B)(A^2 - AB + B^2),$$

$$A^3 - B^3 = (A - B)(A^2 + AB + B^2).$$

Expressions containing negative or fractional powers can also be factored by removing the lowest power common to every term. This technique is useful later when simplifying derivatives and rate equations, even though those expressions are not polynomials.

1.7.1 A factoring decision process

First remove the greatest common factor. Next count the terms. Two terms may form a difference of squares or a sum or difference of cubes. Three terms may form a quadratic trinomial or perfect-square trinomial. Four terms may factor by grouping. After each step, inspect every resulting factor again.

For $-7x^4y^2 + 14xy^3 + 21xy^4$, the greatest common factor is $-7xy^2$. Factoring it out gives

$$-7xy^2(x^3 - 2y - 3y^2).$$

Choosing a negative greatest common factor makes the leading term inside the parentheses positive. Expanding the result verifies the signs.

Worked Example: Removing a common flow factor

Factor $12q^3 - 18q^2$. The greatest common factor is $6q^2$:

$$12q^3 - 18q^2 = 6q^2(2q - 3).$$

Try It. Factor $15r^3 + 10r^2$ completely.

Check Your Work

$$5r^2(3r + 2).$$

For a trinomial with leading coefficient 1, such as $x^2 + 9x - 22$, find two numbers with product -22 and sum 9. The values 11 and -2 give $(x + 11)(x - 2)$. When the leading coefficient is not 1, the split-middle-term method avoids unsupported guessing. In $4x^2 - 8x - 21$, the required product is -84 and the required sum is -8 . The pair 6 and -14 leads to

$$4x^2 + 6x - 14x - 21 = 2x(2x + 3) - 7(2x + 3) = (2x - 7)(2x + 3).$$

The factorization is confirmed by expansion. This last step is short and detects most sign or coefficient errors.

To factor $ax^2 + bx + c$, find two numbers whose product is ac and whose sum is b , split the middle term, and group. For example,

$$6x^2 + 7x + 2 = (3x + 2)(2x + 1).$$

Worked Example: Factoring by grouping

For $6x^2 + 7x + 2$, the product ac is 12. The values 3 and 4 multiply to 12 and add to 7:

$$6x^2 + 3x + 4x + 2 = 3x(2x + 1) + 2(2x + 1) = (3x + 2)(2x + 1).$$

Try It. Factor $x^2 + 9x + 20$.

Check Your Work

$$(x + 4)(x + 5).$$

Worked Example: Using a difference of squares

$$25x^2 - 9 = (5x)^2 - 3^2 = (5x - 3)(5x + 3).$$

Try It. Factor $16y^2 - 49$.

Check Your Work

$$(4y - 7)(4y + 7).$$

Worked Example: Factoring a rectangular-area model

An expanded basin-area model is $A = w^2 + 9w + 20$. Factoring gives

$$A = (w + 4)(w + 5).$$

The factors recover the two dimensions represented by the model.

Try It. Factor the area model $d^2 + 11d + 24$.

Check Your Work

$$(d + 3)(d + 8).$$

Applied Problem: Recovering basin dimensions. A rectangular basin has width $x + 2$ m and length $x + 7$ m. Its expanded area model is $x^2 + 9x + 14$. Factor the polynomial to verify the dimensions, then find the area when $x = 5$.

Check Your Work

$x^2 + 9x + 14 = (x + 2)(x + 7)$. At $x = 5$, the dimensions are 7 m and 12 m, so the area is 84 m².

1.7.2 Practice Problems

- Factor $3x^2 - 6x$.

Check Your Work

$$3x(x - 2).$$

- Factor $x^2 - 11x + 24$.

Check Your Work

$$(x - 3)(x - 8).$$

- Factor $6x^2 + 5x - 4$.

Check Your Work

$$(3x + 4)(2x - 1).$$

- Factor $9a^2 - 25$.

Check Your Work

$$(3a - 5)(3a + 5).$$

- Factor $8x^3 - 1$.

Check Your Work

$$(2x - 1)(4x^2 + 2x + 1).$$

- Factor the pump-expression $18q^2 - 30q$.

Check Your Work

$$6q(3q - 5).$$

- Factor the basin-area model $w^2 + 13w + 40$.

Check Your Work

$$(w + 5)(w + 8).$$

- Factor the difference between two squared pipe diameters, $D^2 - 0.36$.

Check Your Work

$$(D - 0.6)(D + 0.6).$$

1.8 Polynomial division and rational zeros

Higher-degree polynomials require systematic methods for division and for identifying values that make them equal to zero.

Zero of a polynomial. A value c for which $P(c) = 0$.

Division Algorithm for Polynomials. For polynomials P and nonzero D , there are unique polynomials Q and R such that

$$P(x) = D(x)Q(x) + R(x),$$

where $R(x) = 0$ or the degree of R is less than the degree of D .

1.8.1 Long and synthetic division

To divide $6x^2 - 26x + 12$ by $x - 4$, divide leading terms, multiply, subtract, and continue. The result is

$$6x^2 - 26x + 12 = (x - 4)(6x - 2) + 4.$$

The quotient is $6x - 2$ and the remainder is 4. Substitution confirms the remainder because $P(4) = 4$. In synthetic division, include a zero coefficient for every missing power. For $2x^3 - 7x^2 + 5$, use coefficients 2, -7 , 0, 5. Omitting the zero shifts later coefficients into incorrect positions.

Worked Example: Dividing a calibration polynomial

Divide $q^3 - 6q^2 + 11q - 6$ by $q - 1$. Synthetic division with 1 gives quotient $q^2 - 5q + 6$ and remainder 0:

$$q^3 - 6q^2 + 11q - 6 = (q - 1)(q^2 - 5q + 6).$$

Try It. Divide $x^3 - 5x^2 + 6x$ by $x - 2$.

Check Your Work

The quotient is $x^2 - 3x$ and the remainder is 0.

Remainder Theorem. If a polynomial $P(x)$ is divided by $x - c$, the remainder is $P(c)$.

Worked Example: Applying the Remainder Theorem

For $P(x) = 3x^2 - 2x + 5$, the remainder after division by $x - 2$ is

$$P(2) = 3(2)^2 - 2(2) + 5 = 13.$$

Try It. Find the remainder when $P(x) = x^3 + 4x - 1$ is divided by $x + 1$.

Check Your Work

$P(-1) = -1 - 4 - 1 = -6$, so the remainder is -6 .

Worked Example: Finding a remainder by substitution

For $P(x) = 2x^3 - x + 4$, division by $x + 1$ has remainder

$$P(-1) = 2(-1)^3 - (-1) + 4 = 3.$$

Try It. Find the remainder when $x^3 + 2x - 5$ is divided by $x - 2$.

Check Your Work

$$P(2) = 8 + 4 - 5 = 7.$$

Factor Theorem. The expression $x - c$ is a factor of $P(x)$ if and only if $P(c) = 0$.

Worked Example: Testing a proposed factor

For $P(x) = x^3 - 4x^2 + x + 6$,

$$P(2) = 8 - 16 + 2 + 6 = 0.$$

Therefore, $x - 2$ is a factor.

Try It. Determine whether $x + 1$ is a factor of $x^3 + 2x^2 - x - 2$.

Check Your Work

$$P(-1) = -1 + 2 + 1 - 2 = 0, \text{ so } x + 1 \text{ is a factor.}$$

1.8.2 Finding all rational zeros

Rational Zeros Theorem. If $P(x) = a_n x^n + \cdots + a_1 x + a_0$ has integer coefficients and p/q is a rational zero in lowest terms, then p is a factor of a_0 and q is a factor of a_n .

Create the full candidate list, test candidates until one gives remainder zero, divide out the confirmed factor, and repeat. Finish when the remaining quotient is quadratic or factors directly. Finally, multiply the factors to confirm the original polynomial.

For $2x^4 + 3x^3 - 12x^2 - 7x + 6$, the Rational Zeros Theorem limits the search to ratios formed from factors of 6 and factors of 2. Testing a candidate with synthetic division both checks the zero and produces a lower-degree quotient. Repeating the process turns a fourth-degree factorization into smaller problems. The final factors should be multiplied to verify the coefficients.

A zero, a factor, and an intercept express related facts. If $P(c) = 0$, then c is a zero, $x - c$ is a factor, and $(c, 0)$ is an x -intercept of the graph.

Worked Example: Factoring with the Factor Theorem

For $P(x) = x^3 - 7x + 6$, $P(1) = 0$, so $x - 1$ is a factor. Division and further factoring give

$$P(x) = (x - 1)(x + 3)(x - 2).$$

The zeros are -3 , 1 , and 2 .

Try It. Show that $x - 2$ is a factor of $x^3 - 4x^2 + x + 6$.

Check Your Work

$$P(2) = 8 - 16 + 2 + 6 = 0, \text{ so the Factor Theorem confirms it.}$$

Worked Example: Finding all rational zeros of a cubic

For $P(x) = x^3 - 2x^2 - 5x + 6$, testing the rational candidates shows $P(1) = 0$. Dividing by $x - 1$ gives $x^2 - x - 6 = (x - 3)(x + 2)$. Therefore,

$$P(x) = (x - 1)(x - 3)(x + 2),$$

and the rational zeros are -2 , 1 , and 3 .

Try It. Factor $x^3 - x^2 - 4x + 4$ and list its zeros.

Check Your Work

Grouping gives $(x - 1)(x^2 - 4) = (x - 1)(x - 2)(x + 2)$. The zeros are -2 , 1 , and 2 .

Applied Problem: Checking a polynomial sensor model. A dimensionless sensor correction is modelled by $P(x) = x^3 - 4x^2 + x + 6$. Determine whether settings $x = -1$, $x = 2$, and $x = 3$ make the correction zero. Use the results to factor $P(x)$ completely.

Check Your Work

$P(-1) = 0$, $P(2) = 0$, and $P(3) = 0$, so all three settings are zeros. The factorization is $P(x) = (x + 1)(x - 2)(x - 3)$.

1.8.3 Practice Problems

- Find the remainder when $x^2 + 3x + 1$ is divided by $x - 1$.
Check Your Work
 $P(1) = 5$.
- Is $x + 2$ a factor of $x^3 + 3x^2 - 4$?
Check Your Work
Yes, because $P(-2) = 0$.
- List possible rational zeros of $2x^2 - 5x + 2$.
Check Your Work
 $\pm 1, \pm 2, \pm 1/2$.
- Factor $x^3 - 4x^2 - x + 4$.
Check Your Work
 $(x - 4)(x - 1)(x + 1)$.
- Divide $x^2 - 5x + 6$ by $x - 2$.
Check Your Work
Quotient $x - 3$, remainder 0 .
- For a calibration polynomial $P(q) = q^3 - 4q + 3$, find the remainder after division by $q - 1$.
Check Your Work
 $P(1) = 1 - 4 + 3 = 0$, so the remainder is 0 .
- Determine whether $q - 3$ is a factor of $P(q) = q^3 - 5q^2 + 3q + 9$.
Check Your Work
 $P(3) = 27 - 45 + 9 + 9 = 0$, so $q - 3$ is a factor.
- A process model is $P(x) = x^3 - 6x^2 + 11x - 6$. Factor it and list the settings that make $P(x) = 0$.
Check Your Work
 $P(x) = (x - 1)(x - 2)(x - 3)$, so the settings are 1 , 2 , and 3 .

1.9 Equations and an initial look at inequalities

Quantitative problems often state that two expressions represent the same quantity and ask which input makes that statement true.

Equation. A statement that two expressions have equal values.

An equation states that two expressions are equal. Solving means finding every value that makes the original equation true. Whatever operation is applied to one side must also be applied to the other. A solution is checked by substituting it into the original statement. If simplification produces an identity such as $0 = 0$, all values in the original domain solve the equation. If it produces a contradiction such as $0 = 5$, there is no solution.

Linear equations are solved by collecting variable terms on one side and constants on the other. Fractions may be cleared by multiplying every term by the least common denominator. Any values excluded by an original variable denominator remain excluded. The same balancing principle applies to inequalities, except that multiplication or division by a negative number reverses the comparison.

1.9.1 Checking and classifying results

Check a proposed solution by substituting it into both sides of the original statement. An identity such as $0 = 0$ means that every value in the original domain solves the equation. A contradiction such as $0 = 5$ means that no value solves it. For equations with variable denominators, state restrictions before clearing fractions because multiplication can conceal a forbidden input.

For $2x - 7 = 3x + 2$, subtracting $2x$ and then 2 gives $x = -9$. Substitution confirms that both sides equal -25 .

Worked Example: Solving and checking an equation

Solve $2x - 7 = 3x + 2$. Subtracting $2x$ and then 2 gives $x = -9$. The check is

$$2(-9) - 7 = -25 \quad \text{and} \quad 3(-9) + 2 = -25.$$

Try It. Solve and check $5x + 3 = 2x + 18$.

Check Your Work

$3x = 15$, so $x = 5$. Both sides equal 28.

Worked Example: Finding a chemical-feed setting

A feed system delivers 18 L before a variable operating period and then adds 6 L for each hour t . If the required total is 54 L,

$$18 + 6t = 54,$$

so $6t = 36$ and $t = 6$ h. Substitution gives $18 + 6(6) = 54$ L.

Try It. Solve $25 + 8t = 73$ and interpret t as hours.

Check Your Work

$8t = 48$, so $t = 6$ h.

Worked Example: Reversing an inequality

Solve $-3x + 6 > 15$. Subtract 6 to obtain $-3x > 9$. Dividing by -3 reverses the symbol, so $x < -3$.

Try It. Solve $-4x + 1 \leq 13$.

Check Your Work

$-4x \leq 12$, so $x \geq -3$.

Worked Example: Clearing fractions in a detention-time equation

Solve $t/3 + t/6 = 10$. Multiplying every term by 6 gives

$$2t + t = 60,$$

so $3t = 60$ and $t = 20$ min. The original left side is $20/3 + 20/6 = 10$.

Try It. Solve $q/4 + q/12 = 16$.

Check Your Work

Multiplying by 12 gives $3q + q = 192$, so $q = 48$.

Applied Problem: Scheduling tank filling. A tank already contains 12,500 L. Water enters at 850 L/min while 150 L/min leaves through a drain. How many minutes are required for the tank to contain 30,000 L?

Check Your Work

The net inflow is $850 - 150 = 700$ L/min. Solve $12,500 + 700t = 30,000$. Then $700t = 17,500$, so $t = 25$ min.

1.9.2 Practice Problems

1. Solve $4x - 7 = 13$.

Check Your Work

$$x = 5.$$

2. Solve $3(x - 2) = 15$.

Check Your Work

$$x = 7.$$

3. Determine whether $x = 4$ solves $2x + 1 = 9$.

Check Your Work

Yes.

4. Solve $2x + 5 < 11$.

Check Your Work

$$x < 3.$$

5. Solve $-5x \geq 20$.

Check Your Work

$$x \leq -4.$$

6. A tank contains 8,000 L and fills at 600 L/min. Solve $8000 + 600t = 20,000$ for the filling time.

Check Your Work

$$600t = 12,000, \text{ so } t = 20 \text{ min.}$$

7. A dosing system must deliver at least 90 L. It has delivered 30 L and continues at 12 L/min. Solve $30 + 12t \geq 90$.

Check Your Work

$$12t \geq 60, \text{ so } t \geq 5 \text{ min.}$$

8. A process equation is $q/5 + q/10 = 18$. Find q .

Check Your Work

$$\text{Multiplying by 10 gives } 2q + q = 180, \text{ so } q = 60.$$

1.10 Lines and linear applications

When one quantity changes at a constant rate relative to another, a line provides the natural mathematical model.

Slope. The vertical change divided by the horizontal change between two points on a nonvertical line.

The slope through (x_1, y_1) and (x_2, y_2) is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Useful line forms are point-slope form $y - y_1 = m(x - x_1)$, slope-intercept form $y = mx + b$, and general form $Ax + By + C = 0$. Point-slope form is convenient when a point and slope are known. Slope-intercept form displays the constant rate of change and vertical intercept. In general form, set $x = 0$ to find the vertical intercept and $y = 0$ to find the horizontal intercept.

Vertical lines have undefined slope, horizontal lines have slope zero, parallel lines have equal slopes, and perpendicular nonvertical lines have slopes whose product is -1 . Slope has units. If volume is plotted against time, slope is a volume-per-time rate. Its sign indicates whether the response increases or decreases as the input increases.

1.10.1 Intercepts and related lines

The y -intercept is found by setting $x = 0$, and the x -intercept is found by setting $y = 0$. For $y = 5x - 7$, the intercepts are $(0, -7)$ and $(7/5, 0)$. Two distinct nonvertical lines are parallel when their slopes are equal. Two nonvertical lines are perpendicular when their slopes are negative reciprocals. A horizontal and a vertical line are also perpendicular.

A linear model assumes constant change. Its slope must be interpreted with units and within the interval where that assumption is reasonable. Extrapolation far beyond observed values may predict impossible negative populations, volumes, or concentrations, which signals that the linear model should no longer be used.

To compare two lines, rewrite each equation in slope-intercept form unless it is vertical. Equal slopes identify parallel lines. Slopes with product -1 identify perpendicular lines. A line through a specified point and parallel to a known line uses the known slope; a perpendicular line uses its negative reciprocal.

Worked Example: Finding intercepts

For $y = 5x - 7$, setting $x = 0$ gives the y -intercept $(0, -7)$. Setting $y = 0$ gives $5x - 7 = 0$, so the x -intercept is $(7/5, 0)$.

Try It. Find both intercepts of $y = 3x + 6$.

Check Your Work

The intercepts are $(0, 6)$ and $(-2, 0)$.

Worked Example: Modelling tank volume over time

A tank begins with 4.0 m^3 and fills at $0.35 \text{ m}^3/\text{min}$. Its volume after t minutes is

$$V = 0.35t + 4.0.$$

The slope is the inflow rate, and the vertical intercept is the initial volume.

Try It. Write a linear model for a tank beginning at 7 m^3 and draining at $0.25 \text{ m}^3/\text{min}$.
Check Your Work
 $V = -0.25t + 7$.

Worked Example: Temperature conversion

Water freezes at $(0, 32)$ and boils at $(100, 212)$ in Celsius-Fahrenheit coordinates. Thus

$$m = \frac{212 - 32}{100 - 0} = \frac{9}{5},$$

and the conversion equation is

$$F = \frac{9}{5}C + 32.$$

Try It. Use the conversion equation to find the Fahrenheit temperature at $C = 20$.
Check Your Work
 $F = (9/5)(20) + 32 = 68^\circ\text{F}$.

Worked Example: A linear calibration

A sensor gives outputs 2.0 V at 10 mg/L and 5.0 V at 40 mg/L . The slope is $(5 - 2)/(40 - 10) = 0.1 \text{ V per mg/L}$. Using the first point gives $V = 0.1C + 1$.

Try It. Find the output predicted at 25 mg/L .
Check Your Work
 $V = 0.1(25) + 1 = 3.5 \text{ V}$.

Worked Example: Determining a sensor equation from two readings

A turbidity sensor produces 1.4 V at 0.5 NTU and 3.8 V at 2.5 NTU . The slope is

$$m = \frac{3.8 - 1.4}{2.5 - 0.5} = 1.2 \text{ V/NTU}.$$

Using $(0.5, 1.4)$ gives $V - 1.4 = 1.2(T - 0.5)$, or $V = 1.2T + 0.8$.

Try It. A sensor reads 2 V at 10 L/s and 6 V at 30 L/s . Find its linear equation.
Check Your Work
The slope is $(6 - 2)/(30 - 10) = 0.2 \text{ V per L/s}$. Using either point gives $V = 0.2q$.

Applied Problem: Interpreting a declining reservoir model. During a controlled drawdown, reservoir volume is modelled by $V = -1.8t + 72$, where V is in megalitres and t is in hours. Interpret the slope and intercept, find the volume after 15 h , and determine when the model predicts an empty reservoir.

Check Your Work

The intercept 72 ML is the initial volume. The slope -1.8 ML/h is the net loss rate. At 15 h , $V = -1.8(15) + 72 = 45 \text{ ML}$. Setting $V = 0$ gives $t = 40 \text{ h}$.

1.10.2 Practice Problems

- Find the slope through $(1, 2)$ and $(5, 10)$.
Check Your Work
 $m = 2$.
- Find the line with slope 3 through $(2, 1)$.
Check Your Work
 $y = 3x - 5$.
- Find both intercepts of $y = 4x - 8$.
Check Your Work
 $(0, -8)$ and $(2, 0)$.
- Is $y = 2x + 1$ parallel to $2y = 4x - 7$?
Check Your Work
Yes. Both slopes are 2.
- Find a line perpendicular to $y = (1/2)x + 3$ through the origin.
Check Your Work
 $y = -2x$.
- A tank contains 12 m^3 initially and fills at $0.6 \text{ m}^3/\text{min}$. Write its volume model and find the volume after 20 min.
Check Your Work
 $V = 0.6t + 12$; $V(20) = 24 \text{ m}^3$.
- Flow rises from 18 L/s at 2 min to 30 L/s at 8 min. Find the slope and interpret it.
Check Your Work
 $m = (30 - 18)/(8 - 2) = 2 \text{ L/s per min}$. Flow increases by 2 L/s each minute.
- A linear chlorine-sensor model passes through $(0.2, 1.1)$ and $(0.8, 2.9)$, where the input is mg/L and output is volts. Find the model.
Check Your Work
 $m = (2.9 - 1.1)/(0.8 - 0.2) = 3$. Using the first point gives $V = 3C + 0.5$.

1.11 Fractional and rational expressions

Many water-engineering rates and formulas compare one algebraic quantity with another through division.

Fractional expression. A quotient of two algebraic expressions, with all values that make the denominator zero excluded.

Rational expression. A fractional expression whose numerator and denominator are polynomials.

A fractional expression is a quotient of algebraic expressions. It is rational when its numerator and denominator are polynomials. Its domain excludes values that make an original denominator zero. State restrictions before cancelling factors:

$$\frac{x^2 - 16}{x - 4} = x + 4, \quad x \neq 4.$$

Four domain patterns recur. Polynomials allow all real inputs. Rational expressions exclude denominator zeros. An even root requires a nonnegative radicand. An even root in a denominator requires a strictly positive radicand. When several restrictions occur, their intersection is the domain.

Multiplication and division use fraction rules. Factor first and record restrictions before cancelling. When dividing, also exclude values that make the divisor zero before multiplying by its reciprocal. Addition and subtraction require a common denominator. Factor the denominators, construct the least common denominator, rewrite each numerator, combine, and then simplify.

A compound fraction can be cleared by multiplying its entire numerator and denominator by the least common denominator of every smaller fraction. A denominator containing $A + B\sqrt{C}$ can be rationalized using the conjugate $A - B\sqrt{C}$, since their product is $A^2 - B^2C$.

1.11.1 A complete simplification routine

Factor first, state restrictions from every original denominator, perform the required operation, and cancel only common factors. When dividing, values that make the divisor zero must also be excluded. For addition and subtraction, build the least common denominator from each distinct factor at its greatest required power.

For compound fractions, multiply every term in the numerator and denominator by the same least common denominator. For a radical binomial, the conjugate changes the denominator into a difference of squares. These operations change form without changing value, provided all original restrictions are retained.

Since $2 - x = -(x - 2)$, changing the direction of a factor introduces -1 . This is useful when constructing a common denominator, but omitting the negative sign changes the expression. Cancellation applies only to common factors in products. A term in a sum cannot be cancelled.

As an error check, evaluate the original and simplified forms at a convenient permitted value. Matching values do not prove equivalence, but different values immediately reveal an error.

Worked Example: Simplifying while retaining the domain

For $(x^2 - 16)/(x - 4)$, the original denominator requires $x \neq 4$. Factoring and cancelling gives

$$\frac{(x-4)(x+4)}{x-4} = x+4, \quad x \neq 4.$$

The restriction remains because the original expression is undefined at 4.

Try It. Simplify $(x^2 - 25)/(x - 5)$ and state the restriction.

Check Your Work
 $x + 5$, with $x \neq 5$.

Worked Example: Simplifying a hydraulic expression

For $q \neq 5$,

$$\frac{q^2 - 25}{q - 5} = \frac{(q-5)(q+5)}{q-5} = q+5.$$

The simplified form still excludes $q = 5$ because the original expression is undefined there.

Try It. Simplify $(v^2 - 36)/(v - 6)$ and retain the restriction.

Check Your Work
 $v + 6$, with $v \neq 6$.

Worked Example: Multiplying rational expressions

Simplify

$$\frac{3x^2 + 5x + 2}{x - 6} \cdot \frac{10x - 5}{x + 1}.$$

Factoring gives

$$\frac{(3x+2)(x+1)}{x-6} \cdot \frac{5(2x-1)}{x+1} = \frac{5(3x+2)(2x-1)}{x-6},$$

with $x \neq 6, -1$.

Try It. Simplify $[(x^2 - 4)/(x + 3)][3/(x - 2)]$ and state the restrictions.

Check Your Work

$3(x+2)/(x+3)$, with $x \neq -3, 2$.

Worked Example: Adding rational expressions

Using $x(x+1)$ as the common denominator,

$$\frac{2}{x} + \frac{3}{x+1} = \frac{2(x+1) + 3x}{x(x+1)} = \frac{5x+2}{x(x+1)},$$

where $x \neq 0, -1$.

Try It. Add $1/x + 2/(x-1)$.

Check Your Work

$(3x-1)/[x(x-1)]$, with $x \neq 0, 1$.

Worked Example: Combining parallel flow times

If two identical-volume flow paths would individually require t_1 and t_2 hours, the combined reciprocal time is $1/t_1 + 1/t_2$. For $t_1 = 6$ and $t_2 = 4$,

$$\frac{1}{6} + \frac{1}{4} = \frac{2}{12} + \frac{3}{12} = \frac{5}{12}.$$

The corresponding combined time is $12/5 = 2.4$ h.

Try It. Find the combined time when $1/t = 1/3 + 1/6$.

Check Your Work

$1/t = 1/2$, so $t = 2$ h.

Applied Problem: Combining two pump rates. Pump A can fill a tank in 8 h and Pump B can fill it in 12 h. When both pumps operate, their fractional tank-per-hour rates add. Form a rational expression for the combined rate and determine the filling time.

Check Your Work

The combined rate is $1/8 + 1/12 = 3/24 + 2/24 = 5/24$ tank per hour. Therefore, the filling time is the reciprocal, $24/5 = 4.8$ h.

1.11.2 Practice Problems

1. State the domain of $1/(x+4)$.

Check Your Work

$x \neq -4$.

2. Simplify $(x^2 - 9)/(x + 3)$.

Check Your Work

- $x - 3$, with $x \neq -3$.
- Multiply $3/x$ and $2/(x - 1)$.
Check Your Work
 $6/[x(x - 1)]$, with $x \neq 0, 1$.
 - Divide $x/4$ by $3x/8$.
Check Your Work
 $2/3$, with $x \neq 0$.
 - Add $1/x + 1/(x + 2)$.
Check Your Work
 $2(x + 1)/[x(x + 2)]$, with $x \neq 0, -2$.
 - Simplify the hydraulic expression $(q^2 - 16)/(q - 4)$ and state the restriction.
Check Your Work
 $q + 4$, with $q \neq 4$.
 - Two pumps have fractional rates $1/10$ and $1/15$ tank per hour. Find their combined rate and filling time.
Check Your Work
The rate is $1/10 + 1/15 = 1/6$ tank per hour, so the time is 6 h.
 - Simplify $(q^2 - 9)/(q + 3) \cdot 2/(q - 3)$ and state all restrictions.
Check Your Work
The expression simplifies to 2, with $q \neq -3, 3$.

1.12 Solving linear and quadratic equations

A linear equation has the form $ax + b = 0$, with $a \neq 0$. A quadratic has the form $ax^2 + bx + c = 0$, with $a \neq 0$. Quadratics may be solved by factoring or by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The discriminant $D = b^2 - 4ac$ indicates two real solutions when $D > 0$, one repeated real solution when $D = 0$, and no real solutions when $D < 0$. Evaluate it before taking the square root. When substituting negative coefficients into the formula, use parentheses.

1.12.1 Solving by factoring

Factoring is efficient after zero has been placed on one side. The zero-product property then gives one equation from each factor. The square-root property is useful when a squared expression is isolated. Completing the square reveals a perfect square, and the quadratic formula works for every quadratic.

The equation $3x^2 - 10x = 8$ becomes $(3x + 2)(x - 4) = 0$, giving $x = -2/3$ or $x = 4$. Both check algebraically. An application adds a second check: a negative dimension or time is rejected, while a negative signed elevation or rate may be meaningful under a stated convention.

Factoring is efficient only after the equation is written with zero on one side. The zero-product property then states that $AB = 0$ exactly when $A = 0$ or $B = 0$. An equation such as $(x + 2)^2 = 25$ can instead use the square-root property, giving $x + 2 = \pm 5$. In applications, check every algebraic solution against dimensions, signs, and other physical restrictions.

Worked Example: Finding basin dimensions

A basin has area 96 m^2 and length 4 m greater than width w . Then

$$w(w + 4) = 96,$$

$$w^2 + 4w - 96 = (w + 12)(w - 8) = 0.$$

The algebraic candidates are -12 and 8 . Rejecting the negative dimension gives width 8 m and length 12 m.

Try It. A rectangular tank has area 60 m^2 and is 7 m longer than it is wide. Find its dimensions.

Check Your Work

$w(w + 7) = 60$, so $(w + 12)(w - 5) = 0$. The tank is 5 m by 12 m.

Worked Example: Solving a squared depth relationship

If $(d + 1)^2 = 16$, then the square-root property gives

$$d + 1 = \pm 4.$$

Thus $d = 3$ or $d = -5$. If d represents water depth, only $d = 3$ m is physically meaningful.

Try It. Solve $(h - 2)^2 = 25$ and identify the physically meaningful positive depth.

Check Your Work

$h - 2 = \pm 5$, so $h = 7$ or $h = -3$. The meaningful depth is 7 m.

1.12.2 Using the quadratic formula

The quadratic formula is especially useful when a quadratic does not factor conveniently over the integers. First identify a , b , and c from standard form $ax^2 + bx + c = 0$. Substitute signed coefficients with parentheses, simplify the discriminant, and divide the entire numerator by $2a$.

Worked Example: Applying the quadratic formula

For $2x^2 - 3x - 1 = 0$,

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(2)(-1)}}{4} = \frac{3 \pm \sqrt{17}}{4}.$$

Try It. Solve $x^2 - 4x - 1 = 0$.

Check Your Work

$$x = 2 \pm \sqrt{5}.$$

Worked Example: Solving a nonfactorable area equation

Suppose a design equation is $3d^2 + 2d - 7 = 0$. The quadratic formula gives

$$d = \frac{-2 \pm \sqrt{2^2 - 4(3)(-7)}}{2(3)} = \frac{-1 \pm \sqrt{22}}{3}.$$

If d is a depth, retain only $d = (-1 + \sqrt{22})/3 \approx 1.23$.

Try It. Solve $2x^2 + x - 4 = 0$ with the quadratic formula.

Check Your Work

$$x = (-1 \pm \sqrt{33})/4.$$

1.12.3 Interpreting the discriminant

The discriminant is the expression $D = b^2 - 4ac$ under the radical in the quadratic formula. Its sign determines the number of real solutions before the equation is solved: two when $D > 0$, one repeated solution when $D = 0$, and none when $D < 0$.

Worked Example: Interpreting the discriminant

For $x^2 + 3x + 2 = 0$, the discriminant is

$$D = 3^2 - 4(1)(2) = 1 > 0.$$

The equation has two distinct real solutions. Factoring confirms that they are -1 and -2 .

Try It. Use the discriminant to classify the solutions of $x^2 - 6x + 9 = 0$.

Check Your Work

$D = 36 - 36 = 0$, so there is one repeated real solution.

Applied Problem: Designing a rectangular contact basin. A basin must have surface area 180 m^2 . Its length will be 7 m greater than its width. Form and solve a quadratic equation for the dimensions, rejecting any physically impossible result.

Check Your Work

Let the width be w . Then $w(w + 7) = 180$, so $w^2 + 7w - 180 = 0 = (w + 15)(w - 12)$. The candidates are -15 and 12 . Reject the negative value. The basin is 12 m wide and 19 m long.

1.12.4 Practice Problems

- Solve $7x - 9 = 26$.
Check Your Work
 $x = 5$.
- Solve $x^2 - 9 = 0$.
Check Your Work
 $x = \pm 3$.
- Solve $x^2 + 7x + 12 = 0$.
Check Your Work
 $x = -3, -4$.
- Determine the number of real solutions of $3x^2 + 2x + 5 = 0$.
Check Your Work
No real solutions because $D = -56$.
- A square tank floor has area 49 m^2 . Find its positive side length.
Check Your Work
 7 m .
- A rectangular wet well has area 88 m^2 and is 3 m longer than it is wide. Find its dimensions.
Check Your Work
 $w(w + 3) = 88$, so $(w + 11)(w - 8) = 0$. The dimensions are 8 m by 11 m .
- A design equation is $2d^2 - 5d - 3 = 0$. Find the physically meaningful positive solution.
Check Your Work
 $(2d + 1)(d - 3) = 0$, so $d = -1/2$ or 3 . The positive solution is 3 .
- Use the discriminant to determine how many real operating points satisfy $4q^2 - 4q + 5 = 0$.

Check Your Work

$D = (-4)^2 - 4(4)(5) = -64 < 0$, so there are no real operating points.

1.13 Rearranging formulas and radical equations

An established formula may need to be rewritten so that the quantity of interest can be calculated directly.

Equivalent formula. A rearranged equation that expresses the same relationship while isolating a different variable.

To solve a formula for one variable, treat the remaining symbols as constants. From

$$S = 2\pi rh + 2\pi r^2,$$

subtract $2\pi r^2$ and divide by $2\pi r$:

$$h = \frac{S - 2\pi r^2}{2\pi r}, \quad r \neq 0.$$

Worked Example: Isolating height in a cylinder formula

Starting with $S = 2\pi rh + 2\pi r^2$, subtract $2\pi r^2$ and divide by $2\pi r$ to obtain

$$h = \frac{S - 2\pi r^2}{2\pi r}, \quad r \neq 0.$$

Try It. Solve $Q = Av$ for v .

Check Your Work

$v = Q/A$, with $A \neq 0$.

Worked Example: Rearranging the hydraulic residence-time formula

Hydraulic residence time is $t = V/Q$. Solving for flow rate gives

$$tQ = V \quad \Rightarrow \quad Q = \frac{V}{t}, \quad t \neq 0.$$

Try It. Rearrange $L = Qt$ to solve for t .

Check Your Work

$t = L/Q$, with $Q \neq 0$.

Rearranging formulas uses the same equivalent operations as solving numerical equations. Treat all other symbols as constants. If the desired variable occurs in several terms, collect those terms and factor the variable before dividing. Record restrictions created by division, and check the new formula by substituting it into the original relationship.

For a radical equation, determine the domain, isolate the radical, raise both sides to the required power, solve, and check every candidate in the original equation. If another radical remains, isolate it and repeat. The check is essential because squaring can introduce extraneous solutions.

1.13.1 Formula and radical checks

If the desired variable appears in several terms, collect those terms and factor the variable before dividing. For example, $F = GmM/r^2$ becomes $Fr^2 = GmM$, then $m = Fr^2/(GM)$. The final formula should have the units expected for m .

For a radical equation, the domain can eliminate candidates before substitution. In $\sqrt{x+2} = x-4$, the nonnegative left side requires $x \geq 4$. Any solution produced after squaring that is below 4 is extraneous.

Worked Example: Rejecting an extraneous solution

For $\sqrt{x+5} = x-1$, the right side requires $x \geq 1$. Squaring gives $x+5 = x^2-2x+1$, so $(x-4)(x+1) = 0$. Only $x = 4$ satisfies the original equation.

Try It. Solve $\sqrt{x+6} = x$.

Check Your Work

Squaring gives $x = 3$ or -2 ; only $x = 3$ checks.

Worked Example: Solving a radical depth equation

Solve $\sqrt{2d+3} = d$. The right side requires $d \geq 0$. Squaring gives

$$2d + 3 = d^2,$$

so $(d-3)(d+1) = 0$. The candidates are 3 and -1 , but only $d = 3$ satisfies the domain and the original equation.

Try It. Solve $\sqrt{3h+4} = h$.

Check Your Work

Squaring gives $h^2 - 3h - 4 = 0$, so $h = 4$ or -1 . Only $h = 4$ checks.

Applied Problem: Finding flow from detention time. A contact tank has volume 540 m^3 and must provide a detention time of 45 min. Rearrange $t = V/Q$ to solve for Q , then calculate the required flow in cubic metres per minute and litres per second.

Check Your Work

$Q = V/t = 540/45 = 12 \text{ m}^3/\text{min}$. Converting gives $12,000 \text{ L/min}$, then $12,000/60 = 200 \text{ L/s}$.

1.13.2 Practice Problems

- Solve $d = vt$ for t .
Check Your Work
 $t = d/v$, $v \neq 0$.
- Solve $V = LWH$ for H .
Check Your Work
 $H = V/(LW)$, $LW \neq 0$.
- Solve $C = 5(F - 32)/9$ for F .
Check Your Work
 $F = 9C/5 + 32$.
- Solve $\sqrt{x+1} = 3$.
Check Your Work
 $x = 8$.

5. Solve $\sqrt{2x-1} = x-2$.

Check Your Work

The candidates are 1 and 5; only $x = 5$ checks.

6. Rearrange the loading-rate formula $R = Q/A$ to solve for required filter area A .

Check Your Work

$RA = Q$, so $A = Q/R$, with $R \neq 0$.

7. A tank has $V = \pi r^2 h$. Solve for h , then find h when $V = 100 \text{ m}^3$ and $r = 2.5 \text{ m}$.

Check Your Work

$h = V/(\pi r^2)$. Thus $h = 100/[\pi(2.5)^2] \approx 5.09 \text{ m}$.

8. Solve the water-depth equation $\sqrt{2d+8} = d$.

Check Your Work

Squaring gives $d^2 - 2d - 8 = 0$, so $d = 4$ or -2 . Only $d = 4$ checks.

1.14 Solving inequalities

Operating limits, tolerances, and safety constraints usually describe ranges of acceptable values.

Inequality. A comparison between quantities using $<$, $\leq$, $>$, or $\geq$.

Adding or subtracting the same quantity preserves an inequality. Multiplication or division by a positive value also preserves it. Multiplication or division by a negative value reverses it. In a three-part inequality, apply each operation to all three expressions.

For polynomial and rational inequalities, move all terms to one side, combine fractions, factor, identify zeros and excluded values, divide the number line into intervals, determine the sign on each interval, and check endpoint inclusion. A polynomial can change sign only at a zero. A rational expression can also change sign at an excluded denominator zero. Include a numerator zero only when equality is allowed; never include an undefined value.

Worked Example: A quadratic inequality

Solve $x^2 - 12x + 35 < 0$:

$$(x-5)(x-7) < 0.$$

The product is negative between its zeros, so $5 < x < 7$, or $(5, 7)$.

Try It. Solve $(x-2)(x-6) \leq 0$.

Check Your Work

$2 \leq x \leq 6$, or $[2, 6]$.

Worked Example: Solving a compound operating limit

Solve $12 \leq 3q + 6 < 30$. Subtracting 6 from all three expressions and then dividing by 3 gives

$$2 \leq q < 8,$$

or $[2, 8)$.

Try It. Solve $5 < 2q + 1 \leq 13$.

Check Your Work

$$2 < q \leq 6, \text{ or } (2, 6].$$

For $c > 0$, $|x| < c$ means $-c < x < c$, while $|x| > c$ means $x < -c$ or $x > c$. The first statement describes values within c units of zero, while the second describes values farther than c units away. More generally, $|x - a| \leq c$ describes a tolerance of c units around the target a .

1.14.1 Sign charts and tolerances

For $(x-2)(x+3)x(x+5) > 0$, the values $-5, -3, 0, 2$ divide the number line into five intervals. Test one value in each interval and count negative factors. An even number of negative factors gives a positive product. Because the inequality is strict, none of the four zeros is included.

For a rational inequality, numerator zeros and denominator zeros both divide the sign chart, but denominator zeros are always excluded. For $|2x - 3| < 4$, write $-4 < 2x - 3 < 4$ and solve all three parts to obtain $-1/2 < x < 7/2$. A greater-than absolute-value inequality instead produces two outside intervals.

Use a square bracket only when an endpoint is defined and satisfies a non-strict inequality. Use a parenthesis for strict endpoints, undefined values, and infinity. In applications, intersect the algebraic solution with the physical domain. For example, $|q - 20| \leq 3$ gives $[17, 23]$, but an equipment requirement $q \geq 18$ reduces the usable range to $[18, 23]$.

Worked Example: Solving a rational inequality

Solve $1/(x-3) > 0$. The expression is undefined at 3. Its numerator is positive, so the fraction is positive exactly when $x - 3 > 0$. Therefore, $x > 3$, or $(3, \infty)$.

Try It. Solve $1/(x+2) < 0$.

Check Your Work

The denominator must be negative, so $x < -2$, or $(-\infty, -2)$.

Worked Example: Finding a required pump rate

One pump removes no more than 40 L/min for 8.0 minutes. A second pump runs for 5.0 minutes. To remove at least 480 L,

$$320 + 5r \geq 480,$$

so $r \geq 32$ L/min. The second pump must provide at least 32 L/min when the first operates at its maximum rate.

Try It. If 300 L has already been removed, find the rate needed for a second pump operating 6 minutes to reach at least 480 L.

Check Your Work

$300 + 6r \geq 480$, so $r \geq 30$ L/min.

Worked Example: Applying a chlorine-residual tolerance

A residual must remain within 0.15 mg/L of a 0.80 mg/L target. The condition is

$$|C - 0.80| \leq 0.15,$$

so $-0.15 \leq C - 0.80 \leq 0.15$ and therefore $0.65 \leq C \leq 0.95$ mg/L.

Try It. A pH value must remain within 0.25 of 7.2. Find the acceptable interval.

Check Your Work

$$|p - 7.2| \leq 0.25 \text{ gives } 6.95 \leq p \leq 7.45.$$

Applied Problem: Combining process and equipment constraints. A chemical pump can operate from 12 through 40 L/h. A treatment calculation shows that the required rate must satisfy $5r + 20 \geq 145$, while a maximum daily chemical allowance requires $r < 34$ L/h. Find the feasible operating interval.

Check Your Work

The treatment requirement gives $5r \geq 125$, so $r \geq 25$. Combining $r \geq 25$, $r < 34$, and the equipment interval $[12, 40]$ gives $[25, 34)$ L/h.

1.14.2 Practice Problems

- Solve $5x + 13 \leq 7$.
Check Your Work
 $x \leq -6/5$.
- Solve $-2x + 3 > 11$.
Check Your Work
 $x < -4$.
- Solve $(x - 2)(x + 1) \leq 0$.
Check Your Work
 $[-1, 2]$.
- Solve $1/(x - 3) > 0$.
Check Your Work
 $(3, \infty)$.
- Solve $|2x - 1| \geq 5$.
Check Your Work
 $x \leq -2$ or $x \geq 3$.
- A pump rate must be at least 28 L/s but less than 45 L/s. Write the interval.
Check Your Work
 $[28, 45)$ L/s.
- A chlorine residual must stay within 0.10 mg/L of 0.75 mg/L. Write and solve an absolute-value inequality.
Check Your Work
 $|C - 0.75| \leq 0.10$, so $0.65 \leq C \leq 0.85$ mg/L.
- A tank contains 600 L and fills at 35 L/min. How long must it fill to contain at least 1,300 L?
Check Your Work
 $600 + 35t \geq 1300$, so $35t \geq 700$ and $t \geq 20$ min.

1.15 Chapter Review

1.15.1 Chapter summary

Real-number properties, fractions, sets, exponents, and roots establish the language of algebra. Polynomial operations and factoring reveal structure, while polynomial division and zero theorems support further analysis. Equations determine values, lines model constant rates, and inequalities identify ranges. Rational and radical expressions require explicit domain checks. Every applied result should be checked algebraically, dimensionally, and physically.

1.15.2 Common mistakes

- Omitting units or combining incompatible units.
- Confusing display-math delimiters with interval brackets.
- Cancelling terms rather than common factors.
- Dropping domain restrictions after simplification.
- Forgetting to reverse an inequality after division by a negative value.
- Accepting an extraneous or physically impossible answer.

1.15.3 Exercises

1. Write $-4 < x \leq 2.5$ in interval notation.

Check Your Work

$$(-4, 2.5].$$

2. Find the distance between elevations -1.8 m and 4.7 m.

Check Your Work

$$|4.7 - (-1.8)| = 6.5 \text{ m.}$$

3. Simplify x^3x^{-2} and state the original restriction.

Check Your Work

$$x, \text{ with } x \neq 0.$$

4. Factor $4x^2 - 8x - 21$.

Check Your Work

$$(2x - 7)(2x + 3).$$

5. Find the remainder when $2x^3 - 7x^2 + 5$ is divided by $x - 3$.

Check Your Work

$$P(3) = 54 - 63 + 5 = -4.$$

6. Find the line through $(2, 1)$ and $(6, 9)$.

Check Your Work

$$m = 2, \text{ so } y = 2x - 3.$$

7. Simplify $(x^2 - 9)/(x - 3)$ and state its restriction.

Check Your Work

$$x + 3, \text{ with } x \neq 3.$$

8. Solve $3x^2 - 10x = 8$.

Check Your Work

$$x = -2/3 \text{ or } x = 4.$$

9. Solve $4 < 5x - 6 \leq 19$.

Check Your Work

$$2 < x \leq 5, \text{ or } (2, 5].$$

10. A basin has width w , length $w + 5$, and area 84 m^2 . Find its dimensions.

Check Your Work

$w^2 + 5w - 84 = (w + 12)(w - 7) = 0$. Reject $w = -12$. The dimensions are 7 m by 12 m.

1.15.4 Questions for discussion

1. Why should domain restrictions be stated before simplifying a rational expression?
2. When can a negative solution be meaningful in a water-engineering model?
3. How can units reveal an algebra or formula-entry error?

Chapter 2

Functions

Functions provide a precise way to describe how one quantity depends on another. In water engineering, a function may connect time to tank volume, flow to pressure loss, concentration to sensor output, or pipe diameter to cross-sectional area. A function is more than a formula. Its domain states which inputs are permitted, its range states which outputs occur, and its graph shows how the output changes. This chapter follows the progression of the MATH 128 notes from function notation and graphs through rates of change, transformations, combinations, composition, one-to-one and onto behaviour, and inverse functions. Throughout the chapter, units and physical restrictions are treated as part of the model.

Learning outcomes. By the end of this chapter, you should be able to:

- evaluate functions and interpret their inputs and outputs;
- determine domains and ranges from formulas and graphs;
- analyze piecewise functions, net change, and average rate of change;
- solve equations and inequalities graphically and describe graph behaviour;
- graph functions using transformations;
- combine and compose functions while tracking domains; and
- determine whether a function is one-to-one or onto and find an inverse when one exists.

2.1 Function language, evaluation, domain, and range

A function formalizes a dependable input-output relationship. Each allowable input must produce exactly one output, although different inputs may produce the same output.

Function. A rule that assigns exactly one output to every input in its domain.

If a function maps a set A to a set B , write $f : A \rightarrow B$. The set A is the domain and B is the codomain. The value assigned to x is $f(x)$, read “ f of x .” The set of outputs actually produced is the range. The range is contained in the codomain but does not always equal it.

2.1.1 Evaluating functions and recognizing graphs

To evaluate a function, replace every occurrence of its variable with the specified input. Parentheses are essential when the input is negative or algebraic. For $f(x) = x^2 - 3x$, for example,

$$f(-2) = (-2)^2 - 3(-2) = 10.$$

An input a produces the point $(a, f(a))$ on the graph. A graph represents a function of x exactly when every vertical line intersects it at most once. This vertical line test checks whether one input has been assigned more than one output. A circle fails the test, while a vertical parabola such as $y = x^2$ passes.

Worked Example: Evaluating a flow function

The flow through a control setting is modelled by $Q(s) = 2.5s^2 + 4$, where Q is in litres per second. At $s = 3$,

$$Q(3) = 2.5(3)^2 + 4 = 26.5 \text{ L/s.}$$

The ordered pair $(3, 26.5)$ lies on the graph.

Try It. Evaluate $Q(-2)$ for the same algebraic rule.

Check Your Work

$Q(-2) = 2.5(-2)^2 + 4 = 14 \text{ L/s}$. Whether a negative setting is physically permitted is a separate domain question.

Worked Example: Evaluating a difference quotient input

For $f(x) = 1/(x+1)^2$,

$$f(x+h) = \frac{1}{(x+h+1)^2}.$$

The entire expression $x+h$ replaces the original input variable.

Try It. If $g(x) = 3x^2 - 1$, find $g(a+h)$.

Check Your Work

$$g(a+h) = 3(a+h)^2 - 1.$$

2.1.2 Domain, codomain, and range

Domain. The set of inputs for which a function is defined.

Range. The set of outputs a function actually produces.

The algebraic domain is found by excluding operations that are undefined over the real numbers. A denominator cannot equal zero. The radicand of an even root must be nonnegative, and it must be positive when that root appears in a denominator. Logarithmic inputs, introduced later, must be positive. Physical models may add restrictions such as $t \geq 0$, $Q \geq 0$, or a maximum equipment setting.

Range is often easiest to determine from a graph. The function $g(x) = |x|$ has domain $\mathbb{R}$ and range $[0, \infty)$. A quadratic opening upward has a minimum output at its vertex. A reciprocal function may exclude a horizontal value. State range restrictions on outputs, not inputs.

Worked Example: Finding a radical domain

For

$$h(x) = \frac{2+x}{\sqrt{x(x+3)}},$$

the denominator requires $x(x+3) > 0$. A sign chart gives $x < -3$ or $x > 0$, so the domain is $(-\infty, -3) \cup (0, \infty)$.

Try It. Find the domain of $k(x) = 1/\sqrt{x-5}$.

Check Your Work

The denominator requires $x-5 > 0$, so the domain is $(5, \infty)$.

Worked Example: Interpreting the range of a storage model

A tank-depth model is $D(t) = 4 - 0.2t$ for $0 \leq t \leq 20$ min. The depth decreases from 4 m to 0 m. Its physical domain is $[0, 20]$ and its range is $[0, 4]$ m.

Try It. Find the range of $V(t) = 10 + 3t$ for $0 \leq t \leq 8$.

Check Your Work

V increases from 10 to 34, so the range is $[10, 34]$.

Applied Problem: Restricting a reservoir model. Reservoir volume is modelled by $V(t) = 120 - 4.5t$ megalitres, where t is measured in days. Determine the largest physical domain for which time and volume are nonnegative, and state the corresponding range.

Check Your Work

Require $t \geq 0$ and $120 - 4.5t \geq 0$. Thus $0 \leq t \leq 120/4.5 = 26.67$ days. The physical domain is approximately $[0, 26.67]$ days and the range is $[0, 120]$ ML.

2.1.3 Practice Problems

- For $f(x) = x^2 - 4x + 1$, find $f(3)$.
Check Your Work
 $f(3) = 9 - 12 + 1 = -2$.
- If $g(x) = 2/(x-1)$, find $g(a+h)$.
Check Your Work
 $g(a+h) = 2/(a+h-1)$.
- Does $x = y^2$ define y as a function of x over all real values?
Check Your Work
No. For most positive x , the relation has two outputs, $y = \pm\sqrt{x}$.
- Find the domain of $\sqrt{12-3x}$.
Check Your Work
 $12-3x \geq 0$, so $x \leq 4$.
- State the domain and range of $f(x) = (x-2)^2 + 1$.
Check Your Work
Domain $\mathbb{R}$; range $[1, \infty)$.
- A flow model is $Q(v) = 0.8v + 6$. Find $Q(25)$ and interpret it if Q is measured in L/s.
Check Your Work
 $Q(25) = 26$ L/s.
- A water-depth function is $d(t) = 3.6 - 0.12t$ for $0 \leq t \leq 30$. State its range.
Check Your Work
The depth falls from 3.6 m to 0 m, so the range is $[0, 3.6]$ m.

8. Find the physical domain of $A(d) = \sqrt{25 - d^2}$ when d is a nonnegative pipe dimension.

Check Your Work

$$25 - d^2 \geq 0 \text{ and } d \geq 0, \text{ giving } 0 \leq d \leq 5.$$

2.2 Parent functions, piecewise models, and rates of change

Familiar parent graphs make it possible to recognize new functions quickly. Piecewise models allow different rules on different intervals, while rates of change measure how output responds as input changes.

2.2.1 Parent functions and piecewise definitions

Parent function. A basic function whose graph provides a starting shape for a family of related functions.

Important parent functions include constant and linear functions, powers such as x^2 and x^3 , roots such as $\sqrt{x}$ and $\sqrt[3]{x}$, reciprocal functions such as $1/x$ and $1/x^2$, and the absolute-value function $|x|$. For each, know the general shape, domain, range, intercepts, symmetry, and asymptotes.

A piecewise function uses different rules on different parts of its domain. Endpoint symbols determine which rule applies. A solid point represents an included endpoint, while an open point represents an excluded endpoint. Evaluate a boundary input using the rule whose condition includes equality.

Worked Example: Evaluating a piecewise pump rule

Let

$$P(t) = \begin{cases} 12t, & 0 \leq t < 5, \\ 60 + 8(t - 5), & t \geq 5. \end{cases}$$

Then $P(4) = 48$ and $P(5) = 60$. The second rule is used at $t = 5$ because it includes equality.

Try It. Find $P(7)$ for the same function.

Check Your Work

$$P(7) = 60 + 8(7 - 5) = 76.$$

Worked Example: Recognizing an absolute-value response

The function $E(x) = |x - 7|$ measures distance from the target 7. Its graph is V-shaped, has vertex $(7, 0)$, domain $\mathbb{R}$, and range $[0, \infty)$.

Try It. State the vertex and range of $y = |x + 3|$.

Check Your Work

The vertex is $(-3, 0)$ and the range is $[0, \infty)$.

2.2.2 Net change, average rate of change, and difference quotients

Average rate of change. The ratio of net output change to input change over an interval.

Between inputs a and b , net change is $f(b) - f(a)$ and average rate of change is

$$\frac{f(b) - f(a)}{b - a}.$$

Its units are output units per input unit. For inputs a and $a + h$, this becomes the difference quotient

$$\frac{f(a + h) - f(a)}{h}, \quad h \neq 0.$$

For a linear function, the average rate of change is constant and equals its slope. For nonlinear functions, it depends on the interval.

Worked Example: Finding an average inflow rate

A reservoir contains 42 ML at hour 2 and 57 ML at hour 7. The net change is $57 - 42 = 15$ ML, and the average rate is

$$\frac{57 - 42}{7 - 2} = 3 \text{ ML/h.}$$

Try It. Volume rises from 18 m^3 at 4 min to 30 m^3 at 10 min. Find the average rate.

Check Your Work

$$(30 - 18)/(10 - 4) = 2 \text{ m}^3/\text{min.}$$

Worked Example: Simplifying a difference quotient

For $f(x) = x - 3x^2$,

$$\frac{f(2 + h) - f(2)}{h} = \frac{-11h - 3h^2}{h} = -11 - 3h, \quad h \neq 0.$$

Try It. Find $[g(1 + h) - g(1)]/h$ for $g(x) = x^2$.

Check Your Work

$$[(1 + h)^2 - 1]/h = (2h + h^2)/h = 2 + h, \quad h \neq 0.$$

Applied Problem: Comparing filtration rates. Cumulative filtered volume is 18 m^3 after 30 min and 52 m^3 after 80 min. Find the net change and average filtration rate. If the rate were constant, estimate the additional volume filtered during the next 25 min.

Check Your Work

The net change is 34 m^3 . The average rate is $34/(80 - 30) = 0.68 \text{ m}^3/\text{min}$. At that rate, the next 25 min would add $0.68(25) = 17 \text{ m}^3$.

2.2.3 Practice Problems

- State the domain and range of $y = \sqrt{x}$.
Check Your Work
Both are $[0, \infty)$.
- State the domain and range of $y = 1/x$.
Check Your Work
Both are $(-\infty, 0) \cup (0, \infty)$.
- For $f(x) = x + 2$ when $x < 1$ and $f(x) = 3x$ when $x \geq 1$, find $f(1)$.

Check Your Work

$$f(1) = 3.$$

4. Find the average rate of change of x^2 from $x = 1$ to $x = 5$.

Check Your Work

$$(25 - 1)/(5 - 1) = 6.$$

5. Simplify $[(x + h)^2 - x^2]/h$.

Check Your Work

$$2x + h, h \neq 0.$$

6. Tank volume rises from 9 m³ at 3 min to 33 m³ at 15 min. Find the average rate.

Check Your Work

$$(33 - 9)/(15 - 3) = 2 \text{ m}^3/\text{min}.$$

7. A two-stage pump delivers $10t$ L for $0 \leq t < 4$ and $40 + 6(t - 4)$ L for $t \geq 4$. Find the volume at 3 min and 7 min.

Check Your Work

$$P(3) = 30 \text{ L and } P(7) = 58 \text{ L}.$$

8. Concentration changes from 1.8 mg/L at 10 min to 1.2 mg/L at 25 min. Find and interpret the average rate.

Check Your Work

$$(1.2 - 1.8)/(25 - 10) = -0.04 \text{ mg/L per min. Concentration decreases by } 0.04 \text{ mg/L per minute on average.}$$

2.3 Graphical behaviour and transformations

Graphs support both qualitative interpretation and numerical problem solving. They show intersections, sign, intervals of increase or decrease, extrema, and the effects of changing a familiar formula.

2.3.1 Graphical equations, inequalities, and extrema

The solutions of $f(x) = g(x)$ are the x -coordinates where their graphs intersect. The inequality $f(x) < g(x)$ holds where the graph of f lies below the graph of g . Include intersection points only when equality is part of the comparison.

Increasing function. A function whose outputs rise as inputs increase on a specified interval.

Local extremum. A local maximum or minimum value compared with nearby function values.

An interval of increase or decrease is described with input values. A turning point may be a local maximum or minimum. A graph can have several local extrema, and a local extremum need not be the greatest or least value on the entire domain.

Worked Example: Solving an equation graphically

To solve $x^2 - 1 = 8 - x^2$, graph $y = x^2 - 1$ and $y = 8 - x^2$. Their intersections satisfy $2x^2 = 9$, so the intersection inputs are $x = \pm 3/\sqrt{2} \approx \pm 2.12$.

Try It. Find the intersection inputs of $y = x^2$ and $y = 4$.

Check Your Work

$$x = -2 \text{ and } x = 2.$$

Worked Example: Describing a quadratic's behaviour

For $f(x) = -(x-3)^2 + 8$, the vertex $(3, 8)$ is a maximum. The function increases on $(-\infty, 3)$ and decreases on $(3, \infty)$.

Try It. State the increasing and decreasing intervals of $(x+1)^2 - 4$.

Check Your Work

It decreases on $(-\infty, -1)$ and increases on $(-1, \infty)$.

2.3.2 Transforming parent graphs

If $c > 0$, $f(x) + c$ shifts upward and $f(x) - c$ shifts downward. The graph of $f(x - c)$ shifts right, while $f(x + c)$ shifts left. The graph of $-f(x)$ reflects across the x -axis, and $f(-x)$ reflects across the y -axis. Multiplying by a changes vertical scale by $|a|$ and reflects vertically when $a < 0$. Replacing x by kx changes horizontal scale by $1/|k|$.

Order matters. Work from inside to outside: horizontal scale, horizontal shift, vertical scale or reflection, then vertical shift. Track a small set of key points rather than producing a new table from scratch.

Worked Example: Transforming a quadratic

The graph of $y = -3(x+3)^2 + 2$ comes from $y = x^2$. Shift left 3, stretch vertically by 3, reflect across the x -axis, and shift up 2. The vertex is $(-3, 2)$ and the parabola opens downward.

Try It. Describe the transformations and vertex of $y = 2(x-4)^2 - 1$.

Check Your Work

Stretch vertically by 2, shift right 4, and down 1. The vertex is $(4, -1)$.

Worked Example: Transforming a square-root model

The graph of $y = \sqrt{x-3} + 2$ is $y = \sqrt{x}$ shifted right 3 and up 2. Its starting point is $(3, 2)$, its domain is $[3, \infty)$, and its range is $[2, \infty)$.

Try It. State the starting point, domain, and range of $y = -\sqrt{x+1} + 4$.

Check Your Work

Starting point $(-1, 4)$; domain $[-1, \infty)$; range $(-\infty, 4]$.

Applied Problem: Interpreting a transformed pump curve. A simplified pump-efficiency index is $E(q) = -0.02(q-30)^2 + 18$. Identify the transformation from $y = x^2$, state the maximum point, and determine the intervals on which efficiency rises and falls.

Check Your Work

The graph is reflected, vertically compressed by 0.02, shifted right 30, and shifted up 18. Its maximum is $(30, 18)$. It increases for $q < 30$ and decreases for $q > 30$.

2.3.3 Practice Problems

1. Solve graphically or algebraically where $x^2 = 9$.

Check Your Work

$x = -3$ or $x = 3$.

2. Where is $x^2 < 4$?
Check Your Work
 $-2 < x < 2$.
3. State the local minimum of $(x - 5)^2 - 2$.
Check Your Work
The minimum value is -2 at $x = 5$.
4. Describe the transformations from $|x|$ to $-2|x + 1| + 3$.
Check Your Work
Left 1, vertical stretch by 2, reflection across the x -axis, and up 3.
5. State the domain and range of $\sqrt{x + 4} - 2$.
Check Your Work
Domain $[-4, \infty)$; range $[-2, \infty)$.
6. A pump curve is $H(q) = -(q - 20)^2 + 100$. State its maximum and where it decreases.
Check Your Work
Maximum 100 at $q = 20$; decreasing on $(20, \infty)$.
7. Two pressure models are $P_1(q) = 2q + 10$ and $P_2(q) = 70 - q$. Find their intersection.
Check Your Work
 $2q + 10 = 70 - q$, so $q = 20$ and $P = 50$.
8. A transformed depth model is $d(t) = -\sqrt{t - 4} + 6$. State its initial point and physical range for $t \geq 4$.
Check Your Work
It begins at $(4, 6)$. Algebraically the range is $(-\infty, 6]$; restricting to nonnegative depth gives $[0, 6]$.

2.4 Function operations, composition, and inverses

Functions can be combined to build richer models. Algebraic operations combine outputs at the same input, composition passes the output of one function into another, and inversion reverses a one-to-one relationship.

2.4.1 Arithmetic with functions

For functions f and g ,

$$(f + g)(x) = f(x) + g(x), \quad (f - g)(x) = f(x) - g(x),$$

$$(fg)(x) = f(x)g(x), \quad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

The domain of a sum, difference, or product is the intersection of the original domains. A quotient also excludes values for which $g(x) = 0$.

Worked Example: Combining inflow and outflow functions

Let inflow be $I(t) = 40 + 2t$ L/s and outflow be $O(t) = 18 + t$ L/s. Net inflow is

$$(I - O)(t) = 40 + 2t - (18 + t) = 22 + t \text{ L/s.}$$

Try It. If $I(t) = 30 + 3t$ and $O(t) = 12 + 2t$, find net inflow.

Check Your Work

$$(I - O)(t) = 18 + t.$$

2.4.2 Composition of functions

Composite function. The function $(f \circ g)(x) = f(g(x))$, formed by using the output of g as the input of f .

Composition order matters. In general, $f \circ g$ differs from $g \circ f$. Its domain consists of inputs in the domain of g whose outputs also lie in the domain of f .

Worked Example: Composing radius and area

A circular spill radius grows as $r(t) = 24 + 8t$, and area is $A(r) = \pi r^2$. Then

$$(A \circ r)(t) = \pi(24 + 8t)^2.$$

The composition converts time directly to area.

Try It. If a circular tank radius is $r(t) = 2 + 0.1t$, write its surface area as a function of time.

Check Your Work

$$A(t) = \pi(2 + 0.1t)^2.$$

2.4.3 One-to-one and onto functions

One-to-one function. A function in which distinct inputs always produce distinct outputs.

The horizontal line test identifies one-to-one graphs. Algebraically, assume $f(a) = f(b)$ and show that $a = b$. To prove that a function is not one-to-one, one counterexample with two inputs sharing an output is enough.

Onto function. A function whose range equals its stated codomain.

Onto depends on the codomain. The rule $f(x) = x^2$ from $\mathbb{R}$ to $\mathbb{R}$ is not onto because it produces no negative outputs. The same rule from $\mathbb{R}$ to $[0, \infty)$ is onto.

Worked Example: Testing one-to-one behaviour

For $f(x) = 5(x - 2)^5 - 7$, suppose $f(a) = f(b)$. Then $(a - 2)^5 = (b - 2)^5$, so $a - 2 = b - 2$ and $a = b$. Therefore, f is one-to-one.

Try It. Is $h(x) = x^2 - 3x$ one-to-one on $\mathbb{R}$?

Check Your Work

No. For example, $h(0) = 0$ and $h(3) = 0$.

2.4.4 Inverse functions

Inverse function. A function f^{-1} that reverses a one-to-one function, so $f^{-1}(f(x)) = x$ on the domain of f .

The notation $f^{-1}(x)$ does not mean $1/f(x)$. To find an inverse, first establish that f is one-to-one, write $y = f(x)$, interchange x and y , solve for y , and rename the result $f^{-1}(x)$. The domain and range exchange

roles. Verify the result with both compositions.

Worked Example: Inverting a calibration function

For $V(C) = 0.12C + 0.8$, write $y = 0.12C + 0.8$ and solve for C :

$$C = \frac{y - 0.8}{0.12}.$$

Thus $V^{-1}(x) = (x - 0.8)/0.12$, which converts sensor voltage back to concentration.

Try It. Find the inverse of $f(x) = 3x - 5$.

Check Your Work

$$f^{-1}(x) = (x + 5)/3.$$

Applied Problem: Building and reversing a sensor model. A linear sensor converts chlorine concentration C in mg/L to voltage using $V(C) = 1.5C + 0.4$. Find the voltage at 0.8 mg/L, derive the inverse function, and use it to find the concentration corresponding to 2.2 V.

Check Your Work

$V(0.8) = 1.6$ V. Solving $V = 1.5C + 0.4$ gives $C = (V - 0.4)/1.5$, so $V^{-1}(x) = (x - 0.4)/1.5$. At 2.2 V, $C = (2.2 - 0.4)/1.5 = 1.2$ mg/L.

2.4.5 Practice Problems

- If $f(x) = x + 2$ and $g(x) = x^2$, find $(f + g)(x)$.
Check Your Work
 $x^2 + x + 2$.
- For the same functions, find $(f \circ g)(x)$.
Check Your Work
 $x^2 + 2$.
- For the same functions, find $(g \circ f)(x)$.
Check Your Work
 $(x + 2)^2$.
- Is $f(x) = 4x + 1$ one-to-one on $\mathbb{R}$?
Check Your Work
Yes. It is a nonconstant linear function.
- Find the inverse of $f(x) = (2x + 1)/(x - 3)$.
Check Your Work
 $f^{-1}(x) = (3x + 1)/(x - 2)$, with $x \neq 2$.
- Inflow is $I(t) = 50 + 4t$ and outflow is $O(t) = 22 + t$. Find net flow.
Check Your Work
 $I(t) - O(t) = 28 + 3t$ L/s.
- Pipe radius is $r(d) = d/2$ and area is $A(r) = \pi r^2$. Find $(A \circ r)(d)$.
Check Your Work
 $A(d) = \pi(d/2)^2 = \pi d^2/4$.
- A sensor function is $S(C) = 0.25C + 1$. Find its inverse and the concentration at output 3.5.
Check Your Work
 $S^{-1}(x) = 4(x - 1)$. At 3.5, $C = 10$.

2.5 Chapter Review

2.5.1 Chapter summary

A function assigns one output to each input. Domain and range identify allowable inputs and resulting outputs, while graphs reveal shape, change, intersections, and extrema. Parent functions support efficient graphing through transformations. Function arithmetic combines outputs, composition links successive relationships, and inverses reverse one-to-one functions. In applied work, domains, units, and physical meaning remain essential.

2.5.2 Common mistakes

- Substituting a negative or algebraic input without parentheses.
- Confusing codomain with range.
- Ignoring a physical restriction after finding an algebraic domain.
- Reversing the direction of a horizontal shift.
- Assuming that $f \circ g = g \circ f$.
- Treating f^{-1} as a reciprocal.

2.5.3 Exercises

1. For $f(x) = x^2 - 5x + 4$, find $f(-2)$.
Check Your Work
 $f(-2) = 4 + 10 + 4 = 18$.
2. Find the domain of $g(x) = \sqrt{x+2}/(x-4)$.
Check Your Work
 $[-2, 4) \cup (4, \infty)$.
3. Find the average rate of change of $f(x) = x^2 + 1$ from $x = 2$ to $x = 6$.
Check Your Work
 $(37 - 5)/(6 - 2) = 8$.
4. Describe the transformations from x^2 to $-2(x-3)^2 + 5$.
Check Your Work
Vertical stretch by 2, reflection across the x -axis, right 3, and up 5.
5. If $f(x) = 2x - 1$ and $g(x) = \sqrt{x}$, find $(g \circ f)(x)$ and its domain.
Check Your Work
 $\sqrt{2x-1}$ with domain $[1/2, \infty)$.
6. Determine whether $x^3 + 4$ is one-to-one and find its inverse.
Check Your Work
It is one-to-one. The inverse is $f^{-1}(x) = \sqrt[3]{x-4}$.
7. Tank volume is $V(t) = 24 + 1.8t$ m³. Interpret its intercept and slope.
Check Your Work
The tank begins with 24 m³ and gains 1.8 m³ per unit of time.
8. A circular basin radius grows as $r(t) = 5 + 0.02t$. Write area as a function of time.
Check Your Work
 $A(t) = \pi(5 + 0.02t)^2$.

2.5.4 Questions for discussion

1. Why can the physical domain of a water-engineering model be smaller than its algebraic domain?
2. What information can a graph communicate more quickly than a formula?
3. When is an inverse function useful in instrument calibration?

Chapter 3

Polynomial and Rational Functions

Polynomial and rational functions describe curved relationships that linear models cannot capture. Quadratic models represent areas and trajectories, higher-degree polynomials can model responses with several turning points, and rational functions describe ratios whose behaviour may change sharply near excluded inputs. This chapter develops the graphical features emphasized in the MATH 128 notes: vertex form, completing the square, intercepts, local extrema, end behaviour, zeros and multiplicity, asymptotes, holes, and systematic graphing. Water-engineering examples include basin dimensions, pump-response curves, and models whose denominators represent operational limits. Algebraic work is connected continually to graph shape and physical interpretation.

Learning outcomes. By the end of this chapter, you should be able to:

- graph quadratic functions from vertex, standard, and factored forms;
- complete the square and interpret the resulting vertex;
- describe polynomial end behaviour and the effect of zero multiplicity;
- produce a systematic sketch of a polynomial function;
- find domains, intercepts, holes, and asymptotes of rational functions; and
- sketch and interpret rational functions in applied settings.

3.1 Quadratic functions and completing the square

A quadratic function has a parabolic graph whose direction, width, vertex, and intercepts can be read from different algebraic forms.

Quadratic function. A polynomial function $f(x) = ax^2 + bx + c$ with $a \neq 0$.

3.1.1 Vertex form and intercepts

In vertex form,

$$f(x) = a(x - h)^2 + k,$$

the vertex is (h, k) . The graph opens upward when $a > 0$ and downward when $a < 0$. If $a > 0$, the vertex gives the minimum value k ; if $a < 0$, it gives the maximum. The magnitude $|a|$ controls vertical stretch or compression. Find y -intercepts by setting $x = 0$ and x -intercepts by solving $f(x) = 0$.

Worked Example: Graphing from vertex form

For $f(x) = 2(x - 2)^2 - 2$, the vertex is $(2, -2)$ and the parabola opens upward. The y -intercept is $f(0) = 6$. Solving $2(x - 2)^2 - 2 = 0$ gives $x = 1$ and $x = 3$.

Try It. Find the vertex and intercepts of $g(x) = -(x + 1)^2 + 4$.

Check Your Work

Vertex $(-1, 4)$; y -intercept $(0, 3)$; x -intercepts $(-3, 0)$ and $(1, 0)$.

Worked Example: Interpreting a basin-area parabola

A fixed-perimeter design produces $A(w) = -w^2 + 20w$. Factoring gives zeros at 0 and 20, and completing the square gives

$$A(w) = -(w - 10)^2 + 100.$$

The maximum area is 100 m^2 at width 10 m.

Try It. Find the maximum of $A(x) = -x^2 + 16x$.

Check Your Work

$A(x) = -(x - 8)^2 + 64$, so the maximum is 64 at $x = 8$.

3.1.2 Completing the square

Completing the square. Rewriting a quadratic by adding and subtracting the value needed to form a perfect-square trinomial.

For $x^2 + bx$, add and subtract $(b/2)^2$:

$$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2.$$

When $a \neq 1$, first factor a from the variable terms. The vertex of $ax^2 + bx + c$ occurs at $h = -b/(2a)$, with $k = f(h)$.

Worked Example: Completing the square

Rewrite $x^2 + 10x + 7$:

$$x^2 + 10x + 7 = (x + 5)^2 - 25 + 7 = (x + 5)^2 - 18.$$

The vertex is $(-5, -18)$.

Try It. Rewrite $x^2 - 8x + 3$ in vertex form.

Check Your Work

$(x - 4)^2 - 13$.

Worked Example: Completing the square with a leading coefficientFor $2x^2 - 12x + 5$,

$$2(x^2 - 6x) + 5 = 2[(x - 3)^2 - 9] + 5 = 2(x - 3)^2 - 13.$$

The vertex is $(3, -13)$.**Try It.** Rewrite $3x^2 + 12x - 1$ in vertex form.

Check Your Work

$$3(x + 2)^2 - 13.$$

Applied Problem: Maximizing rectangular channel area. A rectangular channel is built inside a 24 m wide lined cross-section so that its width is $24 - 2d$ when its depth is d . Form the area function, rewrite it in vertex form, and find the dimensions that maximize area.

Check Your Work

$A(d) = d(24 - 2d) = -2d^2 + 24d = -2(d - 6)^2 + 72$. The maximum occurs at $d = 6$ m, giving width 12 m and area 72 m².

3.1.3 Practice Problems

- State the vertex of $2(x - 4)^2 + 3$.
Check Your Work
 $(4, 3)$.
- Find the intercepts of $(x - 1)^2 - 9$.
Check Your Work
 y -intercept $(0, -8)$; x -intercepts $(-2, 0)$ and $(4, 0)$.
- Complete the square for $x^2 + 6x - 2$.
Check Your Work
 $(x + 3)^2 - 11$.
- Rewrite $4x^2 - 16x + 1$ in vertex form.
Check Your Work
 $4(x - 2)^2 - 15$.
- Find the maximum or minimum of $-3(x + 2)^2 + 7$.
Check Your Work
Maximum 7 at $x = -2$.
- A basin-area model is $A(w) = -w^2 + 18w$. Find its maximum.
Check Your Work
 $A = -(w - 9)^2 + 81$, so the maximum is 81 m².
- A pump index is $E(q) = -0.5(q - 12)^2 + 40$. State the optimal flow and maximum index.
Check Your Work
Optimal flow 12 units; maximum index 40.
- A rectangular enclosure has area $A(x) = x(30 - 2x)$. Find the positive dimension that maximizes area.
Check Your Work
 $A = -2(x - 7.5)^2 + 112.5$, so $x = 7.5$.

3.2 Polynomial behaviour and graphing

The leading term, zeros, and multiplicities of a polynomial control its large-scale shape. A careful sketch combines these features with intercepts and sign information.

3.2.1 End behaviour and local extrema

End behaviour. The behaviour of a function as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.

For a polynomial $P(x) = a_n x^n + \dots$, the leading term $a_n x^n$ determines end behaviour. Even degree gives matching end directions; odd degree gives opposite directions. A positive leading coefficient points upward as $x \rightarrow \infty$, while a negative one points downward. A degree- n polynomial has at most $n - 1$ local extrema.

Worked Example: Describing cubic end behaviour

For $P(x) = 2x^3 - 5x^2 + x + 1$, the leading term is $2x^3$. Therefore, $P(x) \rightarrow \infty$ as $x \rightarrow \infty$ and $P(x) \rightarrow -\infty$ as $x \rightarrow -\infty$.

Try It. Describe the end behaviour of $-4x^5 + 2x$.

Check Your Work

As $x \rightarrow \infty$, $P(x) \rightarrow -\infty$; as $x \rightarrow -\infty$, $P(x) \rightarrow \infty$.

Worked Example: Bounding the number of turns

A fourth-degree polynomial can have at most three local extrema. A graph showing four distinct turning points therefore cannot represent a fourth-degree polynomial.

Try It. What is the maximum possible number of local extrema for a degree-7 polynomial?

Check Your Work

Six.

3.2.2 Zeros, multiplicity, and a systematic sketch

Multiplicity. The number of times a factor corresponding to a zero occurs in a polynomial.

At a zero of odd multiplicity, a polynomial changes sign and crosses the axis. At a zero of even multiplicity, it touches the axis and turns back. To sketch, factor completely, identify zeros and multiplicities, find the y -intercept, determine signs between zeros, establish end behaviour, and connect the features smoothly.

Worked Example: Sketching a factored cubic

For $P(x) = 5(x + 2)(x - 1)(x - 3)$, the zeros -2 , 1 , and 3 all have multiplicity 1, so the graph crosses at each. The y -intercept is 30. Positive cubic end behaviour runs from lower left to upper right.

Try It. Describe the intercept behaviour of $x(x - 4)^2$.

Check Your Work

It crosses at $x = 0$ and touches without crossing at $x = 4$.

Worked Example: Interpreting repeated process thresholds

For $R(q) = -q^2(2q + 3)(q - 1)$, the zero at $q = 0$ has multiplicity 2, so the graph touches there. It crosses at $q = -3/2$ and $q = 1$. The negative leading term is degree 4, so both ends point downward.

Try It. Describe the zeros and end behaviour of $2(x+1)^2(x-3)$.

Check Your Work

It touches at $x = -1$, crosses at $x = 3$, and has positive odd-degree behaviour from lower left to upper right.

Applied Problem: Analyzing a treatment-response polynomial. A dimensionless response is $R(q) = -0.01q(q-20)^2(q-50)$. Identify the zeros and their multiplicities, state where the graph crosses or touches the axis, and describe its end behaviour.

Check Your Work

The zeros are 0 and 50 with multiplicity 1, and 20 with multiplicity 2. The graph crosses at 0 and 50 and touches at 20. The degree is 4 and the leading coefficient is negative, so both ends approach $-\infty$.

3.2.3 Practice Problems

- Describe the end behaviour of $3x^4 - 2x + 1$.
Check Your Work
Both ends approach ∞ .
- Describe the end behaviour of $-x^3 + 4x$.
Check Your Work
Left end up, right end down.
- State the maximum number of local extrema for degree 6.
Check Your Work
Five.
- Identify zeros and multiplicities of $(x+2)^3(x-5)^2$.
Check Your Work
 -2 has multiplicity 3; 5 has multiplicity 2.
- Find the y -intercept of $2(x-1)(x+3)$.
Check Your Work
 -6 .
- A response model is $P(q) = q(q-10)(q-25)$. At which flows does it cross zero?
Check Your Work
At $q = 0$, 10 , and 25 .
- A model contains $(q-18)^2$. Describe its graph at $q = 18$.
Check Your Work
The zero has even multiplicity, so the graph touches the axis without changing sign.
- Describe the end behaviour of the pump polynomial $-0.2q^4 + 3q^2$.
Check Your Work
Both ends approach $-\infty$.

3.3 Rational functions, holes, and asymptotes

Rational functions are ratios of polynomials. Their graphs can contain excluded points and can approach lines that organize their long-term or near-boundary behaviour.

Vertical asymptote. A line $x = a$ approached by a function whose magnitude grows without bound as x approaches a from at least one side.

3.3.1 Domains, holes, and vertical asymptotes

Factor numerator and denominator before interpreting a rational function. Original denominator zeros are excluded from the domain. A factor that cancels produces a hole, not a vertical asymptote. An uncanceled denominator zero produces a vertical asymptote.

Hole. A removable discontinuity created when a common factor cancels but its zero remains excluded from the original domain.

Worked Example: Distinguishing a hole from an asymptote

For

$$r(x) = \frac{x^2 - 1}{x^2 + x - 2} = \frac{(x-1)(x+1)}{(x-1)(x+2)} = \frac{x+1}{x+2},$$

the domain excludes $x = 1, -2$. There is a hole at $(1, 2/3)$ and a vertical asymptote at $x = -2$.

Try It. Analyze $(x^2 - 4)/(x^2 - x - 6)$ for holes and vertical asymptotes.

Check Your Work

It simplifies to $(x-2)/(x-3)$ with $x \neq -2, 3$. There is a hole at $(-2, 4/5)$ and a vertical asymptote at $x = 3$.

Worked Example: One-sided behaviour near an operating limit

For $f(q) = 1/(q-10)$, $f(q) \rightarrow -\infty$ as $q \rightarrow 10^-$ and $f(q) \rightarrow \infty$ as $q \rightarrow 10^+$. The line $q = 10$ is a vertical asymptote.

Try It. Describe the one-sided behaviour of $1/(x+4)$ near $x = -4$.

Check Your Work

As $x \rightarrow -4^-$, the function approaches $-\infty$; as $x \rightarrow -4^+$, it approaches ∞ .

3.3.2 Horizontal asymptotes

Horizontal asymptote. A line $y = L$ approached by a function as $x \rightarrow \infty$ or $x \rightarrow -\infty$.

For a rational function with numerator degree n and denominator degree m : if $n < m$, the horizontal asymptote is $y = 0$; if $n = m$, it is the ratio of leading coefficients; if $n > m$, there is no horizontal asymptote. A graph may cross a horizontal asymptote.

Worked Example: Finding a horizontal asymptote

For $f(x) = (2x^2 + 7x - 4)/(x^2 + x - 2)$, numerator and denominator have equal degree. The horizontal asymptote is $y = 2/1 = 2$.

Try It. Find the horizontal asymptote of $(3x + 1)/(2x - 5)$.

Check Your Work

$y = 3/2$.

Worked Example: Comparing degrees

For $g(x) = (5x - 1)/(x^3 + 2)$, the numerator degree is less than the denominator degree, so $g(x) \rightarrow 0$ as $x \rightarrow \pm\infty$. The horizontal asymptote is $y = 0$.

Try It. Does $(x^3 + 1)/(x^2 - 4)$ have a horizontal asymptote?

Check Your Work

No. The numerator degree is greater than the denominator degree.

Applied Problem: Interpreting a rational pressure model. A simplified pressure index is $P(q) = (4q^2 - 16)/(q^2 - q - 6)$. Factor the model, state its domain, identify any hole and vertical asymptote, and find the horizontal asymptote.

Check Your Work

$P(q) = 4(q - 2)(q + 2)/[(q - 3)(q + 2)] = 4(q - 2)/(q - 3)$, with $q \neq -2, 3$. The hole is at $(-2, 16/5)$, the vertical asymptote is $q = 3$, and the horizontal asymptote is $P = 4$.

3.3.3 Practice Problems

- State the domain of $(x + 1)/(x^2 - 9)$.
Check Your Work
 $x \neq -3, 3$.
- Find the hole of $(x^2 - 4)/(x - 2)$.
Check Your Work
Hole at $(2, 4)$.
- Find the vertical asymptotes of $(x + 1)/[(x - 2)(x + 5)]$.
Check Your Work
 $x = 2$ and $x = -5$.
- Find the horizontal asymptote of $(7x^2 + 1)/(2x^2 - 3)$.
Check Your Work
 $y = 7/2$.
- Find the horizontal asymptote of $4/(x^2 + 1)$.
Check Your Work
 $y = 0$.
- A head-loss index is $(q^2 - 25)/(q - 5)$. Identify its hole.
Check Your Work
It simplifies to $q + 5$ with a hole at $(5, 10)$.
- A pressure model is $3q/(q - 12)$. Identify its vertical and horizontal asymptotes.
Check Your Work
Vertical $q = 12$; horizontal $P = 3$.
- A treatment model is $(2q + 1)/(q^2 - 16)$. State the excluded flows.
Check Your Work
 $q \neq -4, 4$.

3.4 Sketching and interpreting rational functions

A rational graph is assembled from domain restrictions, discontinuities, intercepts, sign intervals, and end behaviour. Each feature constrains the final sketch.

3.4.1 A systematic graphing process

First factor and state the domain. Record holes before cancelling. Find intercepts, vertical asymptotes, and one-sided behaviour. Use zeros and asymptotes to divide the number line into sign intervals. Determine horizontal behaviour from degrees, then sketch branches that respect every feature.

Worked Example: Graphing a rational function with two asymptotes

For $f(x) = (2x - 1)/(x + 2)$, the domain excludes -2 , the x -intercept is $1/2$, the y -intercept is $-1/2$, the vertical asymptote is $x = -2$, and the horizontal asymptote is $y = 2$. Test points determine the sign of each branch.

Try It. List the key graphing features of $(x + 3)/(x - 1)$.

Check Your Work

Domain $x \neq 1$; x -intercept -3 ; y -intercept -3 ; vertical asymptote $x = 1$; horizontal asymptote $y = 1$.

Worked Example: Graphing with a hole

For $r(x) = (x^2 - 4)/(2x^2 + 2x)$,

$$r(x) = \frac{(x - 2)(x + 2)}{2x(x + 1)}.$$

No factor cancels, so there is no hole. The vertical asymptotes are $x = 0, -1$, zeros are $x = \pm 2$, and the horizontal asymptote is $y = 1/2$.

Try It. Does $(x^2 - 1)/(x^2 + 3x + 2)$ contain a hole?

Check Your Work

Yes. It simplifies from $(x - 1)(x + 1)/[(x + 1)(x + 2)]$ and has a hole at $x = -1$.

3.4.2 Domain and range from a rational graph

Domain is read horizontally and excludes holes and vertical asymptotes. Range is read vertically and may exclude a horizontal value or the output at a hole. Solving $y = f(x)$ for x can help identify excluded output values when the graph is difficult to read.

Worked Example: Finding the range of a reciprocal transformation

For $f(x) = 3 + 2/(x - 4)$, the vertical asymptote is $x = 4$ and the horizontal asymptote is $y = 3$. Since $2/(x - 4)$ can never be zero, the range is $(-\infty, 3) \cup (3, \infty)$.

Try It. Find the domain and range of $-1 + 5/(x + 2)$.

Check Your Work

Domain excludes -2 ; range excludes -1 .

Worked Example: Checking a physically restricted branch

If $H(q) = 40q/(q + 5)$ models head for $q \geq 0$, the algebraic vertical asymptote $q = -5$ lies outside the physical domain. On $q \geq 0$, H begins at 0, increases, and approaches 40 from below.

Try It. Interpret $C(t) = 10t/(t + 2)$ for $t \geq 0$.

Check Your Work

It begins at 0, increases for nonnegative time, and approaches 10 without reaching it.

Applied Problem: Analyzing a saturating treatment model. Removal efficiency is modelled by $E(C) = 90C/(C + 4)$ for dosage $C \geq 0$. Find the intercept, identify the relevant asymptotes, describe the physical range, and calculate $E(8)$.

Check Your Work

The intercept is $(0, 0)$. The algebraic vertical asymptote $C = -4$ is outside the physical domain. The horizontal asymptote is $E = 90$, so the physical range is $[0, 90)$. At $C = 8$, $E = 90(8)/12 = 60$.

3.4.3 Practice Problems

- List the intercepts and asymptotes of $(x - 2)/(x + 1)$.
Check Your Work
 x -intercept 2; y -intercept -2 ; vertical asymptote $x = -1$; horizontal asymptote $y = 1$.
- Determine the sign of $(x - 3)/(x + 2)$ on $(-2, 3)$.
Check Your Work
Negative.
- Find the domain and range of $2 + 1/(x - 5)$.
Check Your Work
Domain excludes 5; range excludes 2.
- Identify all graphing features of $x/(x^2 - 4)$.
Check Your Work
Domain excludes ± 2 ; intercept $(0, 0)$; vertical asymptotes $x = \pm 2$; horizontal asymptote $y = 0$.
- Explain why a graph cannot pass through a hole.
Check Your Work
The corresponding input is excluded from the original domain.
- A dosage response is $E(C) = 75C/(C + 3)$ for $C \geq 0$. Find $E(6)$ and the limiting efficiency.
Check Your Work
 $E(6) = 50$; the limiting efficiency is 75.
- A hydraulic index is $H(q) = 20q/(q + 10)$ for $q \geq 0$. State its physical range.
Check Your Work
 $[0, 20)$.
- A model $P(q) = (q - 5)/(q + 1)$ is used only for $q \geq 0$. Which algebraic features lie outside the physical domain?
Check Your Work
The vertical asymptote $q = -1$ lies outside the physical domain.

3.5 Chapter Review

3.5.1 Chapter summary

Vertex form reveals the turning point of a quadratic, and completing the square converts standard form to vertex form. A polynomial's leading term controls end behaviour, while zeros and multiplicities control axis crossings. Rational functions require careful distinction among holes, vertical asymptotes, and horizontal asymptotes. A reliable graph combines algebraic features, signs, and physical restrictions.

3.5.2 Common mistakes

- Reading the vertex sign incorrectly from $(x - h)^2$.
- Completing the square without accounting for a leading coefficient.

- Assuming every zero produces an axis crossing.
- Treating a cancelled denominator factor as a vertical asymptote.
- Forgetting original domain restrictions.
- Assuming a horizontal asymptote can never be crossed.

3.5.3 Exercises

1. Rewrite $2x^2 + 8x - 3$ in vertex form.
Check Your Work
 $2(x + 2)^2 - 11$.
2. Describe the end behaviour of $-5x^6 + 2x^3$.
Check Your Work
Both ends approach $-\infty$.
3. Describe the zeros of $(x + 3)^2(x - 1)^3$.
Check Your Work
It touches at -3 and crosses at 1 .
4. Find the hole and asymptote of $(x^2 - 9)/(x^2 - x - 6)$.
Check Your Work
Hole at $x = 3$ and vertical asymptote at $x = -2$.
5. Find the horizontal asymptote of $(4x^2 + 1)/(5x^2 - 2x)$.
Check Your Work
 $y = 4/5$.
6. A basin-area model is $A(w) = -2w^2 + 32w$. Find the maximum.
Check Your Work
 $A = -2(w - 8)^2 + 128$, so the maximum is 128 m^2 .

3.5.4 Questions for discussion

1. Which algebraic form of a quadratic is most useful for each graph feature?
2. How can a physically restricted domain change the interpretation of an asymptote?
3. Why should a rational-function sketch be checked against sign intervals?

Chapter 4

Exponential and Logarithmic Functions

Exponential and logarithmic functions describe change governed by multiplication rather than constant addition. They arise in population growth, contaminant decay, compound processes, saturating systems, pH and other logarithmic measurements, and graphs spanning several orders of magnitude. This chapter follows the MATH 128 notes from exponential definitions and transformations through growth models, logarithmic functions, logarithm laws, equation solving, change of base, logarithmic scales, and flow-duration analysis. The emphasis is on translating among formulas, graphs, and practical interpretations while checking domain restrictions and units.

Learning outcomes. By the end of this chapter, you should be able to:

- graph and transform exponential and logarithmic functions;
- construct and interpret growth, decay, compound-interest, and logistic models;
- convert between exponential and logarithmic forms;
- expand and combine logarithmic expressions;
- solve exponential and logarithmic equations and reject invalid solutions;
- use the change-of-base formula and logarithmic scales; and
- interpret semi-log, log-log, and flow-duration graphs.

4.1 Exponential functions and transformations

An exponential function places the variable in an exponent. Its constant multiplicative change distinguishes it from a linear function's constant additive change.

Exponential function. A function $f(x) = ab^x$, where $a \neq 0$, $b > 0$, and $b \neq 1$.

4.1.1 Basic graphs and properties

For $f(x) = b^x$, the domain is $\mathbb{R}$, the range is $(0, \infty)$, the y -intercept is $(0, 1)$, and $y = 0$ is a horizontal asymptote. The function increases when $b > 1$ and decreases when $0 < b < 1$. Equal increases in x multiply the output by the same factor b .

Worked Example: Identifying an exponential base

An exponential graph passes through $(0, 1)$ and $(2, 9)$. If $f(x) = b^x$, then $b^2 = 9$. Since an exponential base is positive, $b = 3$ and $f(x) = 3^x$.

Try It. Find b if $f(x) = b^x$ passes through $(3, 8)$.

Check Your Work

$b^3 = 8$, so $b = 2$.

Worked Example: Interpreting exponential decay

For $C(t) = 12(0.80)^t$, the initial concentration is $C(0) = 12$. Each time unit multiplies concentration by 0.80, which is a 20% decrease.

Try It. Interpret the initial value and percent change in $N(t) = 450(1.06)^t$.

Check Your Work

The initial value is 450 and the quantity grows by 6% per time unit.

4.1.2 Transformations of exponential graphs

The transformation rules developed in the functions chapter apply. In $y = ab^{k(x-h)} + d$, h shifts horizontally, d shifts vertically, a changes vertical scale and may reflect the graph, and k changes horizontal scale. The horizontal asymptote moves from $y = 0$ to $y = d$.

Worked Example: Transforming an exponential graph

For $f(x) = 2(3^{x-1}) + 2$, shift 3^x right 1, stretch vertically by 2, and shift up 2. The horizontal asymptote is $y = 2$, and $f(1) = 4$.

Try It. Describe $g(x) = -2^{x+3} + 1$.

Check Your Work

Shift 2^x left 3, reflect across the x -axis, and shift up 1. The asymptote is $y = 1$.

Worked Example: Determining a transformed range

For $y = 5 - 4(0.5)^x$, the exponential term is positive, so $y < 5$. The domain is $\mathbb{R}$, the range is $(-\infty, 5)$, and the horizontal asymptote is $y = 5$.

Try It. State the range and asymptote of $y = 3(2^x) - 7$.

Check Your Work

Range $(-7, \infty)$; asymptote $y = -7$.

Applied Problem: Modelling contaminant decay. A contaminant concentration is initially 18 mg/L and decreases by 15% each day. Write an exponential model, find the concentration after 6 days, and identify the horizontal asymptote.

Check Your Work

$C(t) = 18(0.85)^t$. Then $C(6) \approx 6.79$ mg/L. The horizontal asymptote is $C = 0$.

4.1.3 Practice Problems

1. State the domain and range of 4^x .
Check Your Work
Domain $\mathbb{R}$; range $(0, \infty)$.
2. Does $(1/3)^x$ increase or decrease?
Check Your Work
It decreases because $0 < 1/3 < 1$.
3. Find the horizontal asymptote of $2(5^x) - 6$.
Check Your Work
 $y = -6$.
4. Find $f(3)$ for $f(x) = 7(0.5)^x$.
Check Your Work
 $7/8 = 0.875$.
5. Write a model with initial value 80 and growth rate 4%.
Check Your Work
 $f(t) = 80(1.04)^t$.
6. Bacterial concentration begins at 240 cells/mL and grows 8% per hour. Write the model.
Check Your Work
 $N(t) = 240(1.08)^t$.
7. Chlorine concentration begins at 2.4 mg/L and retains 92% each hour. Find it after 5 h.
Check Your Work
 $2.4(0.92)^5 \approx 1.58$ mg/L.
8. A transformed treatment model is $R(t) = 60 - 20(0.7)^t$. State its initial value and limiting value.
Check Your Work
 $R(0) = 40$ and the limiting value is 60.

4.2 Exponential models in applications

Exponential models are built from an initial value and a growth factor. Their parameters should be interpreted before calculation so that growth, decay, and units are handled correctly.

4.2.1 Discrete and continuous growth and decay

The form $A(t) = A_0(1 + r)^t$ represents a constant proportional change of r per time period. For decay, r is negative, or use a factor b between 0 and 1. Continuous growth or decay uses

$$A(t) = A_0 e^{kt},$$

where the sign of k determines growth or decay. Compound interest is

$$A(t) = P \left(1 + \frac{r}{n} \right)^{nt},$$

with principal P , annual rate r , n compounding periods per year, and time t in years.

Worked Example: Applying discrete decay

A chemical stock of 50 kg loses 7% per month. After 8 months,

$$A(8) = 50(0.93)^8 \approx 28.0 \text{ kg.}$$

Try It. A quantity begins at 75 and grows 3% per period. Find it after 10 periods.

Check Your Work

$$75(1.03)^{10} \approx 100.8.$$

Worked Example: Continuous growth

For $A(t) = 600e^{0.025t}$, the value after 10 years is

$$A(10) = 600e^{0.25} \approx 770.42.$$

Try It. Find $A(4)$ if $A(t) = 900e^{-0.08t}$.

Check Your Work

$$900e^{-0.32} \approx 653.53.$$

4.2.2 Logistic growth

Logistic model. A growth model that approaches a limiting carrying capacity, commonly written $P(t) = d/(1 + ke^{-ct})$ for positive constants c , d , and k .

The parameter d is the long-term limiting population. At $t = 0$, $P(0) = d/(1 + k)$. Logistic growth is initially close to exponential but slows as limited space or resources become important.

Worked Example: Evaluating a logistic model

For $P(t) = 1200/[1 + 11e^{-0.2t}]$, the initial population is

$$P(0) = \frac{1200}{12} = 100.$$

As $t \rightarrow \infty$, the population approaches 1200.

Try It. Find the initial value and limiting value of $P(t) = 800/[1 + 7e^{-0.3t}]$.

Check Your Work

$P(0) = 800/8 = 100$ and the limiting value is 800.

Worked Example: Comparing two exponential populations

Suppose $A(t) = 1000e^{0.06t}$ and $B(t) = 5000(0.95)^t$. Equality requires

$$1000e^{0.06t} = 5000(0.95)^t.$$

Taking logarithms and solving gives $t = \ln 5/[0.06 - \ln(0.95)] \approx 14.46$ years.

Try It. Solve $200(1.04)^t = 400(0.98)^t$.

Check Your Work

$$t = \ln 2/[\ln(1.04) - \ln(0.98)] \approx 11.66 \text{ periods.}$$

Applied Problem: Comparing decay models. Chlorine concentration is 3.0 mg/L initially. Model A retains 88% each hour, while Model B follows continuous decay $C_B(t) = 3e^{-0.13t}$. Find both concentrations after 8 h and identify which model predicts the larger residual.

Check Your Work

$C_A(8) = 3(0.88)^8 \approx 1.08$ mg/L. $C_B(8) = 3e^{-1.04} \approx 1.06$ mg/L. Model A predicts the slightly larger residual.

4.2.3 Practice Problems

1. Identify the initial value and rate in $A(t) = 420(1.05)^t$.

Check Your Work

Initial value 420; growth rate 5%.

2. Find $A(6)$ for $A(t) = 1000(0.9)^t$.

Check Your Work

531.44.

3. Evaluate $500e^{0.04t}$ at $t = 12$.

Check Your Work

Approximately 808.04.

4. Find the limit of $900/[1 + 5e^{-0.2t}]$ as $t \rightarrow \infty$.

Check Your Work

900.

5. Find the initial value of $600/[1 + 3e^{-0.4t}]$.

Check Your Work

150.

6. A contaminant retains 81% of its mass each day. What percent is lost daily?

Check Your Work

19%.

7. A bacterial count of 350 grows continuously at 9% per hour. Find the count after 5 h.

Check Your Work

$350e^{0.45} \approx 549$.

8. A wetland population follows $P(t) = 2400/[1 + 5e^{-0.1t}]$. Find its initial and limiting populations.

Check Your Work

Initial 400; limiting 2400.

4.3 Logarithmic functions and laws

Logarithms answer exponent questions and provide the inverse operation needed to solve many exponential equations.

4.3.1 Definition, graph, and domain

Logarithm. For $b > 0$ and $b \neq 1$, $\log_b(y) = x$ exactly when $b^x = y$.

The logarithm's input must be positive. The graph of $y = \log_b x$ has domain $(0, \infty)$, range $\mathbb{R}$, x -intercept $(1, 0)$, and vertical asymptote $x = 0$. It is the reflection of $y = b^x$ across $y = x$. Natural logarithm uses base e and is written $\ln x$.

Worked Example: Converting forms

The statement $4^{1/2} = 2$ is equivalent to $\log_4(2) = 1/2$. Likewise, $\log_3(1/9) = -2$ because $3^{-2} = 1/9$.

Try It. Rewrite $5^3 = 125$ in logarithmic form.

Check Your Work

$$\log_5(125) = 3.$$

Worked Example: Finding a logarithmic domain

For $f(x) = \log_7(9 - 5x)$, require $9 - 5x > 0$. Therefore, $x < 9/5$.

Try It. Find the domain of $\ln(3x + 12)$.

Check Your Work

$$3x + 12 > 0, \text{ so } x > -4.$$

4.3.2 Product, quotient, and power laws

For positive arguments,

$$\log_b(MN) = \log_b M + \log_b N,$$

$$\log_b(M/N) = \log_b M - \log_b N,$$

$$\log_b(M^p) = p \log_b M.$$

These laws do not permit splitting a sum or difference inside a logarithm. Domains must be taken from the original expression.

Worked Example: Expanding a logarithm

For positive x and $x + 1$,

$$\ln \left(\frac{3e^x(x+1)^5}{x^2} \right) = \ln 3 + x + 5 \ln(x+1) - 2 \ln x.$$

Try It. Expand $\log_2[8x^3/(x-1)]$.

Check Your Work

$$3 + 3 \log_2 x - \log_2(x-1), \text{ on the original domain.}$$

Worked Example: Combining logarithms

$$3 \ln x + 2 \ln(x+1) - \ln(4x-1) = \ln \left(\frac{x^3(x+1)^2}{4x-1} \right).$$

Try It. Combine $2 \log_5 x - \log_5(x+3)$.

Check Your Work

$$\log_5[x^2/(x+3)].$$

Applied Problem: Determining a pressure parameter. A model gives $\ln(k) = -21.504$, where k is measured in Pa^{-1} . Solve for k and report it in scientific notation.

Check Your Work

Apply the exponential function: $k = e^{-21.504} \approx 4.59 \times 10^{-10} \text{ Pa}^{-1}$.

4.3.3 Practice Problems

1. Rewrite $2^{-4} = 1/16$ logarithmically.
Check Your Work
 $\log_2(1/16) = -4$.
2. Evaluate $\log_{10}(1000)$.
Check Your Work
3.
3. Find the domain of $\ln(7 - 2x)$.
Check Your Work
 $x < 7/2$.
4. Expand $\ln(x^4/y^2)$.
Check Your Work
 $4 \ln x - 2 \ln y$ for positive x, y .
5. Combine $\log x + 3 \log y$.
Check Your Work
 $\log(xy^3)$.
6. A water model gives $\ln k = -8.2$. Find k .
Check Your Work
 $k = e^{-8.2} \approx 2.75 \times 10^{-4}$.
7. Find the domain of $\log(12 - C)$ for concentration $C \geq 0$.
Check Your Work
 $0 \leq C < 12$.
8. Expand $\ln[Q^2/(Q + 5)]$ for positive quantities.
Check Your Work
 $2 \ln Q - \ln(Q + 5)$.

4.4 Solving exponential and logarithmic equations

Inverse properties and logarithm laws allow an unknown exponent or logarithmic argument to be isolated. Every logarithmic candidate must be checked against the original domain.

4.4.1 Exponential equations

If both sides can be written with the same base, equate exponents. Otherwise, isolate the exponential expression, take logarithms, apply the power law, and solve.

Worked Example: Using a common base

Solve $3^{2x+1} = 1$. Since $1 = 3^0$, $2x + 1 = 0$ and $x = -1/2$.

Try It. Solve $2^{3x-2} = 8$.

Check Your Work

$8 = 2^3$, so $3x - 2 = 3$ and $x = 5/3$.

Worked Example: Taking logarithmsFor $3^{2x+3} = 5$,

$$(2x + 3) \ln 3 = \ln 5,$$

so $x = [\ln 5 / \ln 3 - 3] / 2 \approx -0.768$.**Try It.** Solve $4^{x-1} = 17$.

Check Your Work

$$x = 1 + \ln 17 / \ln 4 \approx 3.044.$$

4.4.2 Logarithmic equations and domain checks

Combine logarithms when useful, convert to exponential form, solve the resulting algebraic equation, and reject any value that makes an original logarithmic argument nonpositive.

Worked Example: Solving a logarithmic equation

Solve $\log_7 x + \log_7(x+1) = \log_7(20)$. Then $x(x+1) = 20$, so $(x+5)(x-4) = 0$. The domain requires $x > 0$, leaving $x = 4$.

Try It. Solve $\log_2(x+1) - \log_2(x-1) = 2$.

Check Your Work

 $(x+1)/(x-1) = 4$, so $x = 5/3$, which satisfies $x > 1$.
Worked Example: Solving a decay-time equationIf $C(t) = 3(0.88)^t$ and the target is 1 mg/L, then

$$t = \frac{\ln(1/3)}{\ln(0.88)} \approx 8.59 \text{ h.}$$

Try It. When does $10(0.75)^t = 2$?

Check Your Work

$$t = \ln(0.2) / \ln(0.75) \approx 5.59 \text{ periods.}$$

Applied Problem: Meeting a disinfection residual target. Chlorine residual follows $C(t) = 2.8e^{-0.16t}$ mg/L. Determine when it first falls to 0.50 mg/L and interpret the result.

Check Your Work

$0.50 = 2.8e^{-0.16t}$ gives $t = -\ln(0.50/2.8)/0.16 \approx 10.77$ h. The residual reaches 0.50 mg/L about 10.8 h after the initial measurement.

4.4.3 Practice Problems

1. Solve
- $5^{x+1} = 25$
- .

Check Your Work

$$x = 1.$$

2. Solve
- $2^x = 7$
- .

Check Your Work

$$x = \ln 7 / \ln 2 \approx 2.807.$$

3. Solve $\ln x = 3$.

Check Your Work

$$x = e^3.$$

4. Solve $\log(x) + \log(x - 9) = 1$.

Check Your Work

$$x(x - 9) = 10 \text{ gives } x = 10 \text{ or } -1; \text{ only } x = 10 \text{ is valid.}$$

5. Solve $\ln(2x - 1) = \ln 7$.

Check Your Work

$$2x - 1 = 7, \text{ so } x = 4.$$

6. A contaminant follows $M(t) = 40(0.82)^t$. When is $M = 10$?

Check Your Work

$$t = \ln(0.25) / \ln(0.82) \approx 6.99 \text{ periods.}$$

7. A bacterial count follows $N(t) = 500e^{0.12t}$. When does it reach 1200?

Check Your Work

$$t = \ln(1200/500) / 0.12 \approx 7.30 \text{ h.}$$

8. Solve $\log(C) + \log(C - 3) = \log 28$ for a positive concentration.

Check Your Work

$$C(C - 3) = 28, \text{ so } (C - 7)(C + 4) = 0. \text{ The valid solution is } C = 7.$$

4.5 Change of base, logarithmic scales, and flow duration

Logarithmic scales compress large ranges by representing equal ratios with equal distances. They make multiplicative patterns and power relationships easier to see.

4.5.1 Change of base and orders of magnitude

Change-of-Base Formula. For valid bases a and b ,

$$\log_b x = \frac{\log_a x}{\log_a b}.$$

The number of orders of magnitude separating positive values A and B is $\log_{10}(A/B)$. One order represents a factor of 10.

Worked Example: Changing logarithm base

$$\log_7(20) = \frac{\ln 20}{\ln 7} \approx 1.5395.$$

Try It. Evaluate $\log_3(14)$ using natural logarithms.

Check Your Work

$$\ln 14 / \ln 3 \approx 2.402.$$

Worked Example: Comparing concentration scales

Concentrations 10^5 cells/mL and 10^2 cells/mL differ by

$$\log_{10}(10^5/10^2) = 3$$

orders of magnitude.

Try It. How many orders of magnitude separate 4×10^6 and 4×10^2 ?

Check Your Work

$\log_{10}(10^4) = 4$ orders.

4.5.2 Semi-log, log-log, and flow-duration graphs

Semi-log graph. A graph with one logarithmic axis and one linear axis.

Log-log graph. A graph with logarithmic scales on both axes.

Exponential data become approximately linear when the response is logged and plotted against a linear input. Power-law data become linear when both variables are logged. A flow-duration curve plots discharge against the percentage of time that flow is equalled or exceeded. It helps characterize dependable flow and compare discharge magnitudes across a wide range.

Worked Example: Reading a flow-duration point

If a flow-duration curve shows $18 \text{ m}^3/\text{s}$ at 80% exceedance, then streamflow equals or exceeds $18 \text{ m}^3/\text{s}$ approximately 80% of the observation period.

Try It. Interpret $35 \text{ m}^3/\text{s}$ at 20% exceedance.

Check Your Work

Flow equals or exceeds $35 \text{ m}^3/\text{s}$ about 20% of the time.

Worked Example: Linearizing exponential decay

If $C(t) = C_0 e^{-kt}$, then

$$\ln C = \ln C_0 - kt.$$

A plot of $\ln C$ against t is linear with slope $-k$ and intercept $\ln C_0$.

Try It. Linearize $N(t) = 80e^{0.3t}$.

Check Your Work

$\ln N = \ln 80 + 0.3t$.

Applied Problem: Interpreting dependable streamflow. A flow-duration table reports 42, 25, and $11 \text{ m}^3/\text{s}$ at exceedance percentages 10%, 50%, and 90%. Interpret all three values and identify the median daily flow.

Check Your Work

Flow equals or exceeds $42 \text{ m}^3/\text{s}$ about 10% of the time, $25 \text{ m}^3/\text{s}$ about 50% of the time, and $11 \text{ m}^3/\text{s}$ about 90% of the time. The 50% exceedance value is the median daily flow, $25 \text{ m}^3/\text{s}$.

4.5.3 Practice Problems

1. Evaluate $\log_4(11)$ using change of base.

Check Your Work

- $\ln 11 / \ln 4 \approx 1.730$.
- How many orders separate 10^8 and 10^3 ?
Check Your Work
Five.
 - Which graph often linearizes $y = ab^x$?
Check Your Work
A semi-log graph with a logarithmic y -axis.
 - Which graph often linearizes $y = ax^p$?
Check Your Work
A log-log graph.
 - Interpret 60% exceedance at $14 \text{ m}^3/\text{s}$.
Check Your Work
Flow equals or exceeds $14 \text{ m}^3/\text{s}$ about 60% of the time.
 - Bacterial counts range from 2×10^2 to 2×10^7 cells/mL. How many orders of magnitude separate them?
Check Your Work
Five.
 - A flow-duration curve gives $9 \text{ m}^3/\text{s}$ at 95% exceedance. What does this indicate?
Check Your Work
Flow is at least $9 \text{ m}^3/\text{s}$ approximately 95% of the time.
 - If $C(t) = 6e^{-0.25t}$, state the slope and intercept of a plot of $\ln C$ versus t .
Check Your Work
Slope -0.25 and intercept $\ln 6$.

4.6 Chapter Review

4.6.1 Chapter summary

Exponential models describe constant proportional change, while logarithms recover exponents and invert exponential functions. Logarithm laws support simplification and equation solving, but require positive arguments. Logarithmic scales reveal multiplicative structure, and flow-duration curves summarize how often streamflows meet or exceed selected values.

4.6.2 Common mistakes

- Using the percent rate as the growth factor.
- Forgetting that logarithmic inputs must be positive.
- Splitting the logarithm of a sum.
- Failing to check solutions in the original logarithmic equation.
- Confusing an exceedance percentage with the percentage of observations below a flow.

4.6.3 Exercises

- Find the value after 7 periods for $A(t) = 50(1.08)^t$.
Check Your Work
Approximately 85.69.
- Find the domain of $\ln(4 - 3x)$.
Check Your Work
 $x < 4/3$.
- Expand $\log(x^5/\sqrt{y})$.
Check Your Work
 $5 \log x - (1/2) \log y$.
- Solve $3^x = 20$.
Check Your Work
 $x = \ln 20 / \ln 3 \approx 2.727$.

5. Solve $\ln x + \ln(x - 1) = \ln 6$.

Check Your Work

$$x = 3.$$

6. Interpret a discharge of $16 \text{ m}^3/\text{s}$ at 75% exceedance.

Check Your Work

Discharge equals or exceeds $16 \text{ m}^3/\text{s}$ about 75% of the time.

4.6.4 Questions for discussion

1. What evidence would support choosing an exponential model instead of a linear one?
2. Why are logarithmic axes useful for water-quality measurements spanning a large range?
3. What planning information can a flow-duration curve provide?

Chapter 5

Trigonometric Functions

Trigonometry connects angles, lengths, coordinates, and periodic change. Water Engineering Technology applications include surveying, pipe slopes, tank geometry, channel measurements, rotating equipment, and seasonal or cyclic signals. This chapter follows the MATH 128 notes from angle measure and right triangles through special angles, reference angles, the unit circle, trigonometric graphs, transformations, inverse functions, and identities. Exact values are retained when they clarify structure, while decimal approximations are used when a measured length or angle is required.

Learning outcomes. By the end of this chapter, you should be able to:

- convert angles between degrees and radians and identify coterminal angles;
- solve right triangles using trigonometric ratios;
- evaluate trigonometric functions using reference angles and the unit circle;
- state domains and graph sine, cosine, and tangent functions;
- model periodic behaviour with transformed sine and cosine functions;
- use inverse trigonometric functions to determine angles; and
- verify identities and apply fundamental angle formulas.

5.1 Angles, radians, and right triangles

Angles in standard position begin on the positive x -axis. Positive rotation is counterclockwise and negative rotation is clockwise. Coterminal angles differ by whole revolutions.

5.1.1 Degree and radian measure

Radian. The central angle that intercepts an arc equal in length to the circle's radius.

One revolution is $360^\circ = 2\pi$ radians. Convert degrees to radians by multiplying by $\pi/180^\circ$, and radians to degrees by multiplying by $180^\circ/\pi$.

Worked Example: Converting an angle

$$225^\circ \left(\frac{\pi}{180^\circ} \right) = \frac{5\pi}{4}.$$

Try It. Convert 150° to radians.

Check Your Work

$$5\pi/6.$$

Worked Example: Finding a coterminal angle

$-116^\circ + 360^\circ = 244^\circ$, so 244° is a positive coterminal angle.

Try It. Find a positive coterminal angle for $-5\pi/6$.

Check Your Work

$$-5\pi/6 + 2\pi = 7\pi/6.$$

5.1.2 Right-triangle ratios

For an acute angle θ , $\sin \theta = \text{opposite/hypotenuse}$, $\cos \theta = \text{adjacent/hypotenuse}$, and $\tan \theta = \text{opposite/adjacent}$. Their reciprocals are cosecant, secant, and cotangent.

Worked Example: Recovering all six ratios

If $\sin \theta = 2/5$, take opposite side 2 and hypotenuse 5. The adjacent side is $\sqrt{21}$. Thus $\cos \theta = \sqrt{21}/5$, $\tan \theta = 2/\sqrt{21}$, and the reciprocal ratios follow.

Try It. If $\cos \theta = 3/5$, find $\sin \theta$ and $\tan \theta$ for acute θ .

Check Your Work

The triangle is 3-4-5, so $\sin \theta = 4/5$ and $\tan \theta = 4/3$.

Worked Example: Finding a pipe rise

A 14 m pipe is inclined at 8° . Its vertical rise is

$$14 \sin 8^\circ \approx 1.95 \text{ m.}$$

Try It. Find the rise of a 20 m pipe inclined at 5° .

Check Your Work

$$20 \sin 5^\circ \approx 1.74 \text{ m.}$$

Applied Problem: Surveying a tank. From a point 32 m from a tank, the angle of elevation to its top is 28° . The instrument is 1.6 m above ground. Find the tank height.

Check Your Work

The rise above the instrument is $32 \tan 28^\circ \approx 17.0$ m. Adding 1.6 m gives approximately 18.6 m.

5.1.3 Practice Problems

1. Convert 300° to radians.

Check Your Work

$$5\pi/3.$$

2. Convert $7\pi/12$ to degrees.

Check Your Work

- 105°.
- Find a positive coterminal angle for -70° .
Check Your Work
290°.
 - In a right triangle, opposite = 6 and adjacent = 8. Find $\tan \theta$.
Check Your Work
 $3/4$.
 - Find the hypotenuse when the legs are 5 and 12.
Check Your Work
13.
 - A channel bank rises 2.4 m over a horizontal run of 7.0 m. Find its angle.
Check Your Work
 $\theta = \arctan(2.4/7.0) \approx 18.9^\circ$.
 - A 9 m ladder reaches a tank platform at 65° . Find the vertical reach.
Check Your Work
 $9 \sin 65^\circ \approx 8.16$ m.
 - A pipe falls 0.8 m over 24 m horizontally. Find the downward angle.
Check Your Work
 $\arctan(0.8/24) \approx 1.91^\circ$.

5.2 Special angles, reference angles, and the unit circle

Special triangles and the unit circle provide exact trigonometric values without calculator approximations.

5.2.1 Special triangles and reference angles

The 45° - 45° - 90° triangle has side ratio $1 : 1 : \sqrt{2}$. The 30° - 60° - 90° triangle has ratio $1 : \sqrt{3} : 2$. A reference angle is the acute angle between a terminal side and the x -axis.

Worked Example: Evaluating a special angle

From the 30° - 60° - 90° triangle, $\sin 60^\circ = \sqrt{3}/2$, $\cos 60^\circ = 1/2$, and $\tan 60^\circ = \sqrt{3}$.

Try It. Give the exact values of sine and cosine at 45° .

Check Your Work

Both equal $\sqrt{2}/2$.

Worked Example: Finding a reference angle

The angle 240° lies in quadrant III, so its reference angle is $240^\circ - 180^\circ = 60^\circ$.

Try It. Find the reference angle for $11\pi/6$.

Check Your Work

$2\pi - 11\pi/6 = \pi/6$.

5.2.2 Trigonometric functions of any angle

For a point (x, y) on a terminal side with $r = \sqrt{x^2 + y^2}$, $\sin \theta = y/r$, $\cos \theta = x/r$, and $\tan \theta = y/x$. On the unit circle, the point is $(\cos \theta, \sin \theta)$. Signs follow the quadrant.

Worked Example: Evaluating with a reference angle

315° has reference angle 45° in quadrant IV. Therefore, $\cot 315^\circ = -1$.

Try It. Evaluate $\tan(11\pi/6)$ exactly.

Check Your Work

$$-1/\sqrt{3} = -\sqrt{3}/3.$$

Worked Example: Using a quadrant condition

If $\sin \theta = -1/3$ and $\cos \theta < 0$, then θ is in quadrant III. Using $\cos^2 \theta = 1 - 1/9 = 8/9$ gives $\cos \theta = -2\sqrt{2}/3$ and $\tan \theta = \sqrt{2}/4$.

Try It. If $\cos \theta = -3/5$ in quadrant II, find sine and tangent.

Check Your Work

$$\sin \theta = 4/5 \text{ and } \tan \theta = -4/3.$$

Applied Problem: Resolving a force direction. A 250 N force acts at 135° from the positive horizontal direction. Find its horizontal and vertical components.

Check Your Work

$$F_x = 250 \cos 135^\circ = -125\sqrt{2} \approx -176.8 \text{ N and } F_y = 250 \sin 135^\circ = 125\sqrt{2} \approx 176.8 \text{ N}.$$

5.2.3 Practice Problems

- Evaluate $\sin 30^\circ$.
Check Your Work
 $1/2$.
- Evaluate $\cos(3\pi/4)$.
Check Your Work
 $-\sqrt{2}/2$.
- Find the reference angle for $7\pi/6$.
Check Your Work
 $\pi/6$.
- Evaluate $\sec 240^\circ$.
Check Your Work
 -2 .
- If $(x, y) = (-5, 12)$, find $\sin \theta$.
Check Your Work
 $12/13$.
- A pipe thrust is 80 N at 30° . Find its horizontal component.
Check Your Work
 $80 \cos 30^\circ = 40\sqrt{3} \approx 69.3 \text{ N}$.
- A sloped channel side has direction 210° . State the signs of its horizontal and vertical components.
Check Your Work
Both are negative.
- A survey ray lies in quadrant IV with cosine $5/13$. Find its sine.
Check Your Work
 $-12/13$.

5.3 Trigonometric graphs and transformations

Sine and cosine model smooth periodic change, while tangent models ratios with regularly spaced vertical asymptotes.

5.3.1 Parent graphs and domains

Sine and cosine have domain $\mathbb{R}$, range $[-1, 1]$, and period 2π . Tangent has period π , zeros at $k\pi$, and is undefined at $\pi/2 + k\pi$. Cosecant, secant, and cotangent inherit exclusions from their reciprocal definitions.

Worked Example: Reading the sine graph

Over $[0, 2\pi]$, sine passes through 0 at $0, \pi, 2\pi$, reaches 1 at $\pi/2$, and reaches -1 at $3\pi/2$.

Try It. List the corresponding five key values for cosine.

Check Your Work

1, 0, -1 , 0, 1 at $0, \pi/2, \pi, 3\pi/2, 2\pi$.

Worked Example: Finding tangent exclusions

$\tan x = \sin x / \cos x$ is undefined wherever $\cos x = 0$, so $x = \pi/2 + k\pi$ is excluded.

Try It. Where is $\sec x$ undefined?

Check Your Work

At $x = \pi/2 + k\pi$.

5.3.2 Amplitude, period, and phase shift

For $y = a \sin(kx - b) + d$ or $a \cos(kx - b) + d$, amplitude is $|a|$, period is $2\pi/|k|$, phase shift is b/k , and midline is $y = d$. Apply horizontal changes before vertical changes.

Worked Example: Analyzing a transformed sine curve

For $y = 3 \sin[2(x - \pi/4)]$, amplitude is 3, period is π , phase shift is $\pi/4$ right, and midline is 0.

Try It. Analyze $y = 2 \cos(4x) + 1$.

Check Your Work

Amplitude 2, period $\pi/2$, no phase shift, midline $y = 1$.

Worked Example: Modelling periodic tank depth

A depth varying between 2 m and 6 m has midline 4 and amplitude 2. If its period is 12 h and it begins at a maximum,

$$d(t) = 4 + 2 \cos(\pi t/6).$$

Try It. Write a cosine model with minimum 1, maximum 5, period 8, beginning at a maximum.

Check Your Work

$y = 3 + 2 \cos(\pi t/4)$.

Applied Problem: Modelling daily reservoir depth. Reservoir depth ranges from 7.2 m to 9.8 m with a 24 h period and is highest at midnight. Write a cosine model and find the predicted depth 6 h later.

Check Your Work

Midline 8.5, amplitude 1.3, and angular frequency $2\pi/24 = \pi/12$. Thus $d(t) = 8.5 + 1.3 \cos(\pi t/12)$. At 6 h, $d = 8.5$ m.

5.3.3 Practice Problems

- State the period and range of sine.
Check Your Work
Period 2π ; range $[-1, 1]$.
- State the period of tangent.
Check Your Work
 π .
- Find the amplitude and period of $4 \sin(3x)$.
Check Your Work
Amplitude 4; period $2\pi/3$.
- Find the midline of $-2 \cos x + 5$.
Check Your Work
 $y = 5$.
- Find the phase shift of $\sin[2(x - 1)]$.
Check Your Work
1 unit right.
- Water level ranges from 3 m to 11 m. Find amplitude and midline.
Check Your Work
Amplitude 4 m; midline 7 m.
- A backwash pressure cycle has period 10 min. Find its angular frequency k .
Check Your Work
 $k = 2\pi/10 = \pi/5$ rad/min.
- A depth model is $d(t) = 5 + 1.5 \sin(\pi t/6)$. Find its maximum, minimum, and period.
Check Your Work
Maximum 6.5 m, minimum 3.5 m, period 12 h.

5.4 Inverse trigonometric functions

Sine, cosine, and tangent are restricted to intervals on which they are one-to-one before inverse functions are defined.

5.4.1 Principal values and solving angles

Inverse sine. The function $\arcsin : [-1, 1] \rightarrow [-\pi/2, \pi/2]$ satisfying $\sin(\arcsin x) = x$.

Inverse cosine has range $[0, \pi]$, while inverse tangent has range $(-\pi/2, \pi/2)$. A calculator returns only a principal value. When solving over a larger interval, use reference angles and quadrant signs to find all solutions.

Worked Example: Evaluating an inverse function

$\arcsin(1/2) = \pi/6$, while $\arcsin(-1/2) = -\pi/6$, because both lie in the principal range.

Try It. Evaluate $\arccos(-\sqrt{2}/2)$.

Check Your Work

$3\pi/4$.

Worked Example: Finding all angles in one revolution

If $\cos \theta = -0.2250$ for $0 \leq \theta < 360^\circ$, the reference angle is approximately 77.0° . Cosine is negative in quadrants II and III, so $\theta \approx 103.0^\circ$ or 257.0° .

Try It. Solve $\sin \theta = 0.6$ for $0 \leq \theta < 360^\circ$.

Check Your Work

$\theta \approx 36.87^\circ$ or 143.13° .

Worked Example: Evaluating a composition exactly

If $\alpha = \arcsin(9/11)$, then a right triangle gives adjacent side $\sqrt{40}$ and hypotenuse 11. Therefore,

$$\cos(\arcsin(9/11)) = \frac{2\sqrt{10}}{11}.$$

Try It. Evaluate $\tan[\arccos(3/5)]$.

Check Your Work

$4/3$.

Worked Example: Finding a channel angle

A channel rises 1.2 m over 18 m horizontally. Its inclination is

$$\theta = \arctan(1.2/18) \approx 3.81^\circ.$$

Try It. Find the inclination for a 0.9 m rise over 25 m.

Check Your Work

$\arctan(0.9/25) \approx 2.06^\circ$.

Applied Problem: Locating two possible pipe directions. A pipe direction satisfies $\sin \theta = -0.35$ for $0 \leq \theta < 360^\circ$. Find both directions to the nearest tenth of a degree.

Check Your Work

The reference angle is $\arcsin(0.35) \approx 20.5^\circ$. Sine is negative in quadrants III and IV, giving 200.5° and 339.5° .

5.4.2 Practice Problems

1. Evaluate $\arcsin(0)$.

Check Your Work

0.

2. Evaluate $\arccos(1/2)$.

Check Your Work

$\pi/3$.

3. Evaluate $\arctan(1)$.
Check Your Work
 $\pi/4$.
4. Solve $\cos \theta = 0.4$ on $0 \leq \theta < 360^\circ$.
Check Your Work
 $\theta \approx 66.4^\circ, 293.6^\circ$.
5. Evaluate $\sin(\arccos(5/13))$.
Check Your Work
 $12/13$.
6. A 2.5 m rise occurs over 40 m. Find the slope angle.
Check Your Work
 $\arctan(2.5/40) \approx 3.58^\circ$.
7. A tank roof rises 3 m over a half-span of 8 m. Find its angle.
Check Your Work
 $\arctan(3/8) \approx 20.6^\circ$.
8. A force has vertical component 45 N and magnitude 120 N. Find possible directions in one revolution.
Check Your Work
 $\sin \theta = 0.375$, giving approximately 22.0° and 158.0° .

5.5 Trigonometric identities and angle formulas

Identities are equations true for every input in their common domain. They allow expressions to be rewritten and exact values to be calculated.

5.5.1 Fundamental identities

Pythagorean Identities. Wherever defined,

$$\sin^2 \theta + \cos^2 \theta = 1, \quad \tan^2 \theta + 1 = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta.$$

Quotient identities are $\tan \theta = \sin \theta / \cos \theta$ and $\cot \theta = \cos \theta / \sin \theta$. To verify an identity, transform one side into the other using valid algebra rather than manipulating both sides toward an unknown middle.

Worked Example: Rewriting tangent

If θ is in quadrant III and $\cos \theta = c < 0$, then $\sin \theta = -\sqrt{1 - c^2}$ and

$$\tan \theta = \frac{\sqrt{1 - c^2}}{-c}.$$

Try It. Express $\sin \theta$ in terms of positive $\cos \theta = c$ in quadrant IV.

Check Your Work

$$\sin \theta = -\sqrt{1 - c^2}.$$

Worked Example: Verifying an identity

$$\frac{\cos \theta}{1 - \sin \theta} \cdot \frac{1 + \sin \theta}{1 + \sin \theta} = \frac{1 + \sin \theta}{\cos \theta} = \sec \theta + \tan \theta.$$

Try It. Verify $\cos \theta (\sec \theta - \cos \theta) = \sin^2 \theta$.

Check Your Work

The left side is $1 - \cos^2 \theta = \sin^2 \theta$.

5.5.2 Even, odd, addition, and subtraction formulas

Sine and tangent are odd, while cosine is even. The addition formulas include

$$\sin(s + t) = \sin s \cos t + \cos s \sin t,$$

$$\cos(s + t) = \cos s \cos t - \sin s \sin t.$$

Replace t by $-t$ for subtraction formulas.

Worked Example: Finding an exact compound angle

Using $75^\circ = 45^\circ + 30^\circ$,

$$\sin 75^\circ = \frac{\sqrt{2}}{2} \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Try It. Find $\cos 75^\circ$ exactly.

Check Your Work

$(\sqrt{6} - \sqrt{2})/4$.

Worked Example: Using odd symmetry

$\sin(-40^\circ) = -\sin 40^\circ$, while $\cos(-40^\circ) = \cos 40^\circ$.

Try It. Simplify $\tan(-x)$.

Check Your Work

$-\tan x$.

Applied Problem: Combining two directional changes. A survey direction is formed by adding 45° and 30° . Use angle formulas to find its exact sine and cosine, then confirm that their squares sum to 1.

Check Your Work

$\sin 75^\circ = (\sqrt{6} + \sqrt{2})/4$ and $\cos 75^\circ = (\sqrt{6} - \sqrt{2})/4$. Squaring and adding gives 1.

5.5.3 Practice Problems

1. Simplify $1 - \sin^2 x$.
Check Your Work
 $\cos^2 x$.
2. Rewrite $\tan x$ using sine and cosine.
Check Your Work
 $\sin x / \cos x$.
3. Verify $\sec^2 x - \tan^2 x = 1$.
Check Your Work
It is a rearrangement of $1 + \tan^2 x = \sec^2 x$.
4. Find $\cos 15^\circ$ exactly.
Check Your Work

- $(\sqrt{6} + \sqrt{2})/4$.
5. Simplify $\cos(-x)$.
Check Your Work
 $\cos x$.
6. If a pipe direction is -25° , relate its sine to $\sin 25^\circ$.
Check Your Work
 $\sin(-25^\circ) = -\sin 25^\circ$.
7. A resultant direction is $105^\circ = 60^\circ + 45^\circ$. Find its exact cosine.
Check Your Work
 $(\sqrt{2} - \sqrt{6})/4$.
8. If $\sin \theta = 3/5$ in quadrant II, use an identity to find tangent.
Check Your Work
 $\cos \theta = -4/5$, so $\tan \theta = -3/4$.

5.6 Chapter Review

5.6.1 Chapter summary

Degree and radian measures describe the same rotations. Right-triangle ratios, reference angles, and the unit circle extend trigonometry to all angles. Sine and cosine model periodic behaviour through amplitude, period, phase shift, and midline. Inverse functions recover principal angles, while identities and angle formulas support exact rewriting and verification.

5.6.2 Common mistakes

- Using the wrong calculator angle mode.
- Assigning the wrong sign after finding a reference angle.
- Confusing inverse trigonometric functions with reciprocals.
- Reporting only the principal value when an interval requires multiple angles.
- Applying a horizontal transformation in the wrong direction.

5.6.3 Exercises

1. Convert $11\pi/9$ to degrees.
Check Your Work
 220° .
2. Find all six ratios for a 5-12-13 triangle.
Check Your Work
Relative to the angle opposite 5: $\sin = 5/13$, $\cos = 12/13$, $\tan = 5/12$ and reciprocals.
3. Evaluate $\sin(5\pi/4)$.
Check Your Work
 $-\sqrt{2}/2$.
4. Analyze $y = -3\cos(2x) + 4$.
Check Your Work
Amplitude 3, period π , midline 4, reflected.
5. Solve $\cos \theta = -0.6$ for one revolution.
Check Your Work
 $\theta \approx 126.9^\circ, 233.1^\circ$.
6. Verify $(1 - \cos^2 x)/\sin x = \sin x$ where defined.
Check Your Work
 $1 - \cos^2 x = \sin^2 x$, then divide by $\sin x$.

5.6.4 Questions for discussion

1. When is an exact trigonometric value preferable to a decimal approximation?
2. How do domain restrictions differ for tangent and inverse tangent?
3. What physical assumptions are needed before using a sinusoidal water-level model?

Chapter 6

Statistics

Statistics turns measurements into evidence. Water professionals collect flow, pressure, turbidity, concentration, rainfall, and equipment data, then summarize patterns and judge whether the observations support a practical conclusion. This chapter develops statistics as a full component of the book. It follows the MATH 128 sequence from data matrices and study design through numerical summaries, variability, quartiles, boxplots, outliers, skewness, and the choice of appropriate measures. The central distinction is between describing the data that were observed and making claims that the data-collection method can support. The statistical framework is consistent with the OpenIntro material used in the source notes (Diez, Barr, and Çetinkaya-Rundel 2019).

Learning outcomes. By the end of this chapter, you should be able to:

- identify cases, variables, and variable types in a data matrix;
- distinguish populations, samples, parameters, and statistics;
- recognize sampling bias and compare sampling methods;
- distinguish observational studies from experiments and evaluate design principles;
- calculate and interpret measures of centre and variability;
- construct five-number summaries and identify potential outliers; and
- describe distribution shape and select appropriate numerical summaries.

6.1 Data, variables, populations, and samples

Statistical work begins by deciding what each row represents, what has been measured, and which larger group the study is intended to describe.

6.1.1 Data matrices and variable types

Data matrix. A rectangular arrangement in which rows represent cases and columns represent variables.

Variable. A recorded characteristic that can differ from one case to another.

Numerical variables arise from measurement or counting, so arithmetic such as differences and averages can be meaningful. Categorical variables place cases into groups or labels. Numerical variables may be continuous, such as flow, or discrete, such as the number of pump failures.

Worked Example: Reading a water-quality data matrix

Suppose each row represents a sampling date and columns record site, turbidity, temperature, and alarm status. Sampling date is the case identifier, site and alarm status are categorical variables, and turbidity and temperature are numerical variables.

Try It. Classify pipe material, pipe diameter, and break count by variable type.

Check Your Work

Material is categorical, diameter is numerical continuous, and break count is numerical discrete.

Worked Example: Identifying cases and variables

A table contains 60 rows, each representing one filter run, and columns for run time, terminal head loss, and media type. There are 60 cases and three measured variables.

Try It. A file has one row for each of 24 wells and columns for depth, yield, and aquifer class. Identify cases and variables.

Check Your Work

The 24 wells are cases. Depth and yield are numerical; aquifer class is categorical.

6.1.2 Populations, samples, parameters, and statistics

Population. The complete collection of cases about which a study seeks information.

Sample. The subset of the population from which data are collected.

A parameter describes a population, while a statistic is calculated from a sample. The sample mean $\bar{x}$ estimates the population mean μ . A useful research question identifies the target population, variables, and time or setting of interest.

Worked Example: Identifying a target population

To estimate average summer turbidity in a watershed, technicians collect weekly samples at selected sites. The population is all relevant summer turbidity measurements in the watershed, and the collected readings form the sample.

Try It. Identify the population and sample when 40 of 600 service connections are tested for lead.

Check Your Work

The 600 connections form the population; the 40 tested connections form the sample.

Worked Example: Distinguishing a statistic from a parameter

The mean of 30 sampled chlorine residuals is 0.72 mg/L. This value is a statistic. The unknown mean residual across all relevant times and locations is a parameter.

Try It. Is the failure rate calculated from every pump owned by a facility a statistic or parameter?

Check Your Work

It is a parameter for that complete facility pump population.

Applied Problem: Designing a monitoring table. A utility wants to compare raw and treated water at 12 dates. For each date it records source, turbidity before treatment, turbidity after treatment, temperature, and whether an alarm occurred. Identify the cases and classify every variable.

Check Your Work

The 12 dates are cases. Source and alarm status are categorical. Both turbidity measurements and temperature are numerical continuous variables.

6.1.3 Practice Problems

1. Define a case in a data matrix.
Check Your Work
The observational unit represented by one row.
2. Classify colour as numerical or categorical.
Check Your Work
Categorical.
3. Classify the number of valve failures.
Check Your Work
Numerical discrete.
4. Distinguish $\bar{x}$ from μ .
Check Your Work
 $\bar{x}$ is a sample mean; μ is a population mean.
5. Why is a sample used?
Check Your Work
Measuring an entire population may be impractical, costly, destructive, or impossible.
6. A sample contains 25 daily flow values from a year. Identify population and sample.
Check Your Work
Population: all daily flows in the year; sample: the 25 selected days.
7. Classify pH, treatment plant, and complaint count.
Check Your Work
pH numerical continuous; plant categorical; complaint count numerical discrete.
8. The mean pressure of every hydrant in a network is calculated. Is it a statistic or parameter?
Check Your Work
A parameter for that network.

6.2 Sampling, observational studies, and experiments

The way data are collected determines which conclusions are defensible. A large biased sample can be less useful than a smaller well-designed sample.

6.2.1 Bias and sampling methods

Sampling bias. A systematic tendency for a sampling method to overrepresent or underrepresent parts of the population.

Convenience samples select cases that are easy to reach. Non-response bias occurs when respondents differ systematically from nonrespondents. Simple random sampling gives each case an equal selection opportunity.

Stratified sampling draws from each relatively similar subgroup. Cluster sampling selects whole mixed groups, while multistage sampling combines selection stages.

Worked Example: Recognizing convenience bias

Testing only taps nearest the treatment plant may not represent the full distribution network. Distance, pipe age, and residence time may differ elsewhere, creating convenience sampling bias.

Try It. Why might a voluntary online water-taste survey be biased?

Check Your Work

People with unusually strong opinions may be more likely to respond.

Worked Example: Choosing a stratified sample

To compare pipe conditions across three material types, divide connections into material strata and randomly sample within each. This ensures that every material is represented.

Try It. Suggest strata for sampling water quality across a city with four pressure zones.

Check Your Work

Use the four pressure zones as strata and randomly select locations within each.

6.2.2 Observational studies and experiments

Observational study. A study that records naturally occurring variables without assigning treatments.

Observational studies can establish association but not causation because confounding variables may explain the pattern. Experiments assign treatments and can support causal conclusions when well designed. Core principles are comparison, random assignment, replication, and blocking. Blinding reduces behavioural and measurement effects.

Worked Example: Distinguishing study types

Comparing historical filter records for two coagulants is observational if operators chose the coagulant. Randomly assigning comparable filter runs to the two coagulants would create an experiment.

Try It. Is recording rainfall and overflow events observational or experimental?

Check Your Work

Observational. Rainfall is not assigned by the researchers.

Worked Example: Blocking a treatment experiment

If raw-water temperature may affect removal, group test runs into low- and high-temperature blocks, then randomly assign coagulant doses within each block.

Try It. Suggest a blocking variable for testing pump efficiency across old and new pumps.

Check Your Work

Pump age or model could define blocks before treatments are randomized.

Applied Problem: Evaluating a filter-media study. A facility installs new media in its newest filter and compares its turbidity with an older filter using standard media. Can the result establish that media caused any difference? Identify a confounder and propose a stronger design.

One Possible Answer

No. Filter age or condition is confounded with media type. A stronger design would use several comparable filters, randomly assign media where feasible, replicate runs, and block by relevant conditions such as raw-water turbidity.

6.2.3 Practice Problems

1. Define convenience sampling.
Check Your Work
Selecting cases mainly because they are easy to reach.
2. What is simple random sampling?
Check Your Work
A random method giving each population case an equal selection opportunity.
3. Distinguish strata from clusters.
Check Your Work
Strata are internally similar and sampled within; clusters should each resemble the population and selected clusters are sampled.
4. Can an observational association prove causation?
Check Your Work
No.
5. Name the four experiment principles.
Check Your Work
Comparison, randomization, replication, and blocking.
6. Sampling only daytime flows creates what concern?
Check Your Work
Time-of-day sampling bias.
7. A utility randomly samples within each pressure zone. Name the method.
Check Your Work
Stratified sampling.
8. Operators randomly assign two coagulant doses within high- and low-turbidity blocks. Identify randomization and blocking.
Check Your Work
Dose is randomly assigned; raw-water turbidity defines the blocks.

6.3 Measures of centre and variability

Centre describes a typical value, while variability describes how widely observations differ. Both are needed to characterize numerical data.

6.3.1 Mean, median, and mode

Sample mean. The arithmetic average $\bar{x} = (\sum_{i=1}^n x_i)/n$.

The median is the middle ordered value, or the average of the two middle values when n is even. The mode is the most frequent value. Mean uses every observation and is sensitive to extremes; median depends mainly on order and is resistant.

Worked Example: Calculating centre

For turbidity values 0.20, 0.24, 0.25, 0.31, 0.80, the mean is $1.80/5 = 0.36$ NTU and the median is 0.25 NTU. The high value pulls the mean upward.

Try It. Find mean and median of 2, 3, 3, 4, 8.

Check Your Work

Mean 4; median 3.

Worked Example: Updating a mean

Five readings have mean 12, so their sum is 60. Adding a sixth reading of 18 gives new mean $(60 + 18)/6 = 13$.

Try It. Four values have mean 9. Add a fifth value of 14 and find the new mean.

Check Your Work

Original sum 36; new mean $50/5 = 10$.

6.3.2 Variance and standard deviation

Sample standard deviation. The square root of sample variance,

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}.$$

Variance uses squared deviations and has squared units. Standard deviation returns to the original measurement units and describes typical distance from the mean. Population variance uses divisor N instead of $n - 1$.

Worked Example: Computing sample variability

For 2, 4, 6, the mean is 4. Squared deviations are 4, 0, and 4. Thus $s^2 = 8/(3 - 1) = 4$ and $s = 2$.

Try It. Find the sample standard deviation of 5, 5, 5.

Check Your Work

All deviations are zero, so $s = 0$.

Worked Example: Comparing consistency

Two sensors both average 1.0 mg/L. Sensor A has $s = 0.03$ mg/L and Sensor B has $s = 0.18$ mg/L. Sensor A's readings are more tightly clustered around their mean.

Try It. Which is more consistent: $s = 2.1$ kPa or $s = 0.8$ kPa?

Check Your Work

The readings with $s = 0.8$ kPa.

Applied Problem: Comparing two turbidity periods. Period A readings are 0.20, 0.25, 0.30 NTU. Period B readings are 0.10, 0.25, 0.40 NTU. Calculate each mean and sample standard deviation, then compare centre and consistency.

Check Your Work

Both means are 0.25 NTU. For A, $s = 0.05$ NTU. For B, $s = 0.15$ NTU. The periods have equal centres, but A is more consistent.

6.3.3 Practice Problems

- Find the mean of 4, 6, 8.
Check Your Work
6.
- Find the median of 2, 5, 7, 11.
Check Your Work
6.
- Find the mode of 1, 2, 2, 4.
Check Your Work
2.
- Find sample variance and standard deviation of 1, 3, 5.
Check Your Work
 $s^2 = 4$, $s = 2$.
- Which has the same units as the data, variance or standard deviation?
Check Your Work
Standard deviation.
- Find the mean flow of 18, 20, 22, and 24 L/s.
Check Your Work
21 L/s.
- Chlorine readings are 0.6, 0.6, 0.7, 0.8, 2.1. Which centre is more resistant?
Check Your Work
The median, 0.7 mg/L.
- Two pressure datasets have equal mean, with standard deviations 4 kPa and 11 kPa. Which is more variable?
Check Your Work
The dataset with 11 kPa.

6.4 Quartiles, outliers, and distribution shape

Ordered summaries show how data are distributed and provide measures that remain useful when observations are skewed or include extremes.

6.4.1 Five-number summaries and boxplots

Interquartile range. The spread of the middle half of the data, $IQR = Q_3 - Q_1$.

The five-number summary is minimum, Q_1 , median, Q_3 , maximum. A boxplot displays these values. Potential outliers lie below $Q_1 - 1.5IQR$ or above $Q_3 + 1.5IQR$. A flagged point should be investigated, not automatically deleted.

Worked Example: Finding outlier fences

If $Q_1 = 12$ and $Q_3 = 20$, then $IQR = 8$. The fences are $12 - 12 = 0$ and $20 + 12 = 32$. Values below 0

or above 32 are potential outliers.

Try It. Find fences when $Q_1 = 4$ and $Q_3 = 10$.

Check Your Work

$IQR = 6$; fences are -5 and 19 .

Worked Example: Constructing a five-number summary

For 2, 4, 5, 7, 9, 12, 15, the median is 7, $Q_1 = 4$, and $Q_3 = 12$. The five-number summary is (2, 4, 7, 12, 15).

Try It. Find the five-number summary of 1, 3, 4, 6, 8.

Check Your Work

Using medians of the halves excluding the overall median: (1, 2, 4, 7, 8).

6.4.2 Skewness and choosing summaries

Skewed distribution. A distribution with a longer tail on one side than the other.

Right-skewed distributions usually have mean greater than median; left-skewed distributions usually have mean less than median. For roughly symmetric data without strong outliers, use mean and standard deviation. For skewed data or data with outliers, use median and IQR.

Worked Example: Choosing resistant summaries

Most repair times are under 4 h, but one major failure requires 38 h. The distribution is right-skewed, so median and IQR better describe a typical repair time and spread.

Try It. Which summaries suit a symmetric set with no outliers?

Check Your Work

Mean and standard deviation.

Worked Example: Predicting the effect of an extreme value

Increasing the largest observation from 64,401 to 150,000 raises the mean and standard deviation substantially. The median and IQR may remain unchanged because the ordered middle positions do not move.

Try It. What happens to the mean if an unusually large value is added?

Check Your Work

It generally increases, often noticeably.

Applied Problem: Summarizing filter run times. Run times in hours are 18, 19, 20, 20, 21, 22, 23, 24, 25, 52. Describe the likely shape and choose appropriate measures of centre and spread. Explain the effect of 52.

One Possible Answer

The value 52 creates a long right tail, so the distribution is right-skewed. Median and IQR are preferable. The value 52 raises the mean and standard deviation much more than it affects the median and IQR and should be investigated as a possible unusual operating event.

6.4.3 Practice Problems

1. Name the five values in a five-number summary.
Check Your Work
Minimum, Q_1 , median, Q_3 , maximum.
2. Find the IQR if $Q_1 = 14$ and $Q_3 = 29$.
Check Your Work
15.
3. Find the upper fence for $Q_1 = 10$, $Q_3 = 18$.
Check Your Work
 $IQR = 8$, so upper fence = 30.
4. Which is more resistant, mean or median?
Check Your Work
Median.
5. For right skew, which is usually larger, mean or median?
Check Your Work
Mean.
6. Pressure data contain one sensor error of 900 kPa among values near 400 kPa. Which summaries should be reported initially?
Check Your Work
Median and IQR, with the extreme value investigated.
7. Flow quartiles are 12 and 28 L/s. Find IQR and outlier fences.
Check Your Work
 $IQR = 16$; the fences are -12 L/s and 52 L/s.
8. Repair times have a long right tail. Choose centre and spread.
Check Your Work
Median and IQR.

6.5 Chapter Review

6.5.1 Chapter summary

A data matrix organizes cases and variables. Samples provide statistics that estimate population parameters, but sampling methods determine representativeness. Observational studies establish association, while randomized experiments can support causal conclusions. Mean and standard deviation suit symmetric data; median and IQR resist skewness and outliers. Potential outliers should be investigated in context.

6.5.2 Common mistakes

- Treating an identifier as a measured variable.
- Generalizing from a convenience sample without qualification.
- Claiming causation from an observational association.
- Using the population variance formula for a sample.
- Removing a potential outlier without investigation.
- Reporting mean and standard deviation for strongly skewed data without explanation.

6.5.3 Exercises

1. Classify flow rate and pipe material.

Check Your Work

Flow rate is numerical continuous; pipe material is categorical.

2. Explain why sampling only one neighbourhood may be biased.

One Possible Answer

The neighbourhood may differ from others in pipe age, pressure zone, or water residence time.

3. Find mean and median of 3, 4, 4, 5, 19.

Check Your Work

Mean 7; median 4.

4. Find sample standard deviation of 2, 4, 6.

Check Your Work

2.

5. Find fences when $Q_1 = 20$ and $Q_3 = 32$.

Check Your Work

$IQR = 12$; fences 2 and 50.

6. Choose summaries for strongly right-skewed overflow volumes.

Check Your Work

Median and IQR.

6.5.4 Questions for discussion

1. What practical conditions can make a water-quality sample unrepresentative?
2. When would a randomized water-treatment experiment be unethical or impractical?
3. Why should an apparent outlier be investigated before it is removed?

Glossary

Absolute value. The distance of a real number from zero on the number line.

Average rate of change. The ratio of net output change to input change over an interval.

Completing the square. Rewriting a quadratic by adding and subtracting the value needed to form a perfect-square trinomial.

Composite function. The function $(f \circ g)(x) = f(g(x))$, formed by using the output of g as the input of f .

Data matrix. A rectangular arrangement in which rows represent cases and columns represent variables.

Domain. The set of inputs for which a function is defined.

End behaviour. The behaviour of a function as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.

Equation. A statement that two expressions have equal values.

Equivalent formula. A rearranged equation that expresses the same relationship while isolating a different variable.

Exponent. A number that indicates the power to which a base is raised.

Exponential function. A function $f(x) = ab^x$, where $a \neq 0$, $b > 0$, and $b \neq 1$.

Factor. An expression multiplied by another expression to produce a given product.

Fraction. A quotient a/b in which a is the numerator, b is the denominator, and $b \neq 0$.

Fractional expression. A quotient of two algebraic expressions, with all values that make the denominator zero excluded.

Function. A rule that assigns exactly one output to every input in its domain.

Hole. A removable discontinuity created when a common factor cancels but its zero remains excluded from the original domain.

Horizontal asymptote. A line $y = L$ approached by a function as $x \rightarrow \infty$ or $x \rightarrow -\infty$.

Increasing function. A function whose outputs rise as inputs increase on a specified interval.

Inequality. A comparison between quantities using $<$, $\leq$, $>$, or $\geq$.

Interquartile range. The spread of the middle half of the data, $IQR = Q_3 - Q_1$.

Interval. A continuous set of real numbers between specified endpoints.

Inverse function. A function f^{-1} that reverses a one-to-one function, so $f^{-1}(f(x)) = x$ on the domain of f .

Inverse sine. The function $\arcsin : [-1, 1] \rightarrow [-\pi/2, \pi/2]$ satisfying $\sin(\arcsin x) = x$.

Local extremum. A local maximum or minimum value compared with nearby function values.

Logarithm. For $b > 0$ and $b \neq 1$, $\log_b(y) = x$ exactly when $b^x = y$.

Logistic model. A growth model that approaches a limiting carrying capacity, commonly written $P(t) = d/(1 + ke^{-ct})$ for positive constants c , d , and k .

Log-log graph. A graph with logarithmic scales on both axes.

Multiplicity. The number of times a factor corresponding to a zero occurs in a polynomial.

Observational study. A study that records naturally occurring variables without assigning treatments.

One-to-one function. A function in which distinct inputs always produce distinct outputs.

Onto function. A function whose range equals its stated codomain.

Parent function. A basic function whose graph provides a starting shape for a family of related functions.

Polynomial. A finite sum of terms whose variable exponents are nonnegative integers.

Population. The complete collection of cases about which a study seeks information.

Quadratic function. A polynomial function $f(x) = ax^2 + bx + c$ with $a \neq 0$.

Radian. The central angle that intercepts an arc equal in length to the circle's radius.

Range. The set of outputs a function actually produces.

Rational expression. A fractional expression whose numerator and denominator are polynomials.

Rational number. A number that can be written as a/b , where a and b are integers and $b \neq 0$.

Real number. A number represented by a point on the real number line. The set of all real numbers is denoted by $\mathbb{R}$.

Root. A value that produces a given number when raised to a specified power.

Sample. The subset of the population from which data are collected.

Sample mean. The arithmetic average $\bar{x} = (\sum_{i=1}^n x_i)/n$.

Sample standard deviation. The square root of sample variance, $s = \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 / (n - 1)}$.

Sampling bias. A systematic tendency for a sampling method to overrepresent or underrepresent parts of the population.

Semi-log graph. A graph with one logarithmic axis and one linear axis.

Set. A collection of distinct objects called elements.

Skewed distribution. A distribution with a longer tail on one side than the other.

Slope. The vertical change divided by the horizontal change between two points on a nonvertical line.

Variable. A recorded characteristic that can differ from one case to another.

Vertical asymptote. A line $x = a$ approached by a function whose magnitude grows without bound as x approaches a from at least one side.

Zero of a polynomial. A value c for which $P(c) = 0$.

Appendix A

Formula Reference

This appendix collects formulas used repeatedly in the book. Each formula should be used with the domain restrictions, units, and assumptions developed in its chapter.

A.1 Algebra and geometry

$$V_{\text{rectangular}} = LWH, \quad A_{\text{circle}} = \pi r^2, \quad V_{\text{cylinder}} = \pi r^2 h.$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad D = b^2 - 4ac.$$

A.2 Functions and rates

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad \text{average rate} = \frac{f(b) - f(a)}{b - a}.$$

$$(f \circ g)(x) = f(g(x)).$$

A.3 Exponential and logarithmic models

$$A(t) = A_0(1 + r)^t, \quad A(t) = A_0 e^{kt},$$

$$A(t) = P \left(1 + \frac{r}{n} \right)^{nt}, \quad P(t) = \frac{d}{1 + ke^{-ct}}.$$

$$\log_b(MN) = \log_b M + \log_b N,$$

$$\log_b(M/N) = \log_b M - \log_b N,$$

$$\log_b(M^p) = p \log_b M, \quad \log_b x = \frac{\ln x}{\ln b}.$$

A.4 Trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$

$$180^\circ = \pi \text{ radians}, \quad a^2 + b^2 = c^2.$$

$$\sin^2 \theta + \cos^2 \theta = 1,$$

$$\sin(s + t) = \sin s \cos t + \cos s \sin t,$$

$$\cos(s + t) = \cos s \cos t - \sin s \sin t.$$

A.5 Descriptive statistics

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}, \quad s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}, \quad s = \sqrt{s^2}.$$

$$IQR = Q_3 - Q_1,$$

$$\text{lower fence} = Q_1 - 1.5IQR, \quad \text{upper fence} = Q_3 + 1.5IQR.$$

References

Diez, David M., Christopher D. Barr, and Mine Çetinkaya-Rundel. 2019. *OpenIntro Statistics*. 4th ed. OpenIntro.

